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Tactical asset allocation with macroeconomic regime detection

D. C. OLIVEIRA [†], D. SANDFELDER ^{‡*}, A. FUJITA ^{†§}, X. DONG [‡] and M. CUCURINGU^{‡¶||}

[†]Institute of Mathematics and Statistics, University of São Paulo, São Paulo, Brazil

[‡]Oxford-Man Institute of Quantitative Finance, University of Oxford, Oxford, UK

[§]Medical Institute of Bioregulation, Kyushu University, Fukuoka, Japan

[¶]Department of Mathematics, University of California Los Angeles, Los Angeles, CA, USA

^{||}Department of Statistics and Oxford-Man Institute of Quantitative Finance, University of Oxford, Oxford, UK

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This paper extends the literature on tactical asset allocation by incorporating regime modeling using techniques from machine learning. We propose a novel model that classifies current regimes, forecasts the distribution of future regimes, and integrates these forecasts with the historical performance of individual assets to optimize portfolio allocations. Using a macroeconomic data set from the FRED-MD database, our approach employs a modified k-means algorithm to ensure consistent regime classification over time. We then leverage these regime predictions to estimate expected returns and volatilities, which are subsequently mapped into portfolio allocations using various sizing schemes. Our method outperforms traditional benchmarks, including equal-weight, buy-and-hold, and random regime models. Furthermore, we are the first to apply a regime detection model from a large macroeconomic dataset to tactical asset allocation, demonstrating significant improvements in portfolio performance. Our work presents several key contributions, including a novel data-driven regime detection algorithm tailored to uncertainty in forecasted regimes, as well as the application of the FRED-MD data set for tactical asset allocation.

Keywords: Tactical allocation; Regime detection; Market analysis; Time series; Macroeconomic data

1. Introduction

The asset allocation literature primarily concerns building optimal portfolios tailored to the utility functions of the investors. The most popular framework for asset allocation and portfolio choice is the seminal work of Markowitz (1952). Known as mean-variance optimization (MVO), this method estimates expected returns and covariances for financial assets. Then, it uses these estimates to solve an optimization problem to determine portfolio weights. This method is widely used in practice to maximize returns for a given level of portfolio risk.

The tactical asset allocation literature closely follows Markowitz's work, focusing on signals that can be used as tactical portfolio tilts to mitigate tail-risk scenarios. A typical tactical asset allocation model consists of three steps: (1)

building signals to model the state of the economy and/or assets, (2) using the time series of asset prices and/or the signals constructed in the previous step to develop individual asset forecasts, and (3) mapping these forecasts into portfolio weights or allocations. Among the various signals used for tactical asset allocation, the outputs of the models that account for the dynamics of the regime are particularly notable, as shown in Ang and Bekaert (2004) and Guidolin and Timmermann (2007). These models aim to estimate probabilities for a given number of unobserved states (regimes) using observed data.

The regime modeling literature is extensive and has been explored from both econometric and machine learning perspectives. From an econometric perspective, the seminal work of Hamilton (1989) laid the foundation for subsequent studies. This strand of literature has primarily focused on regime-switching models, typically with a specified number of estimated states. In contrast, the machine learning

*Corresponding author. Email: dylan.sandfelder@eng.ox.ac.uk

literature is mostly based on clustering techniques, as in Chen *et al.* (2021) and Prakash *et al.* (2021). A recent study by Chen *et al.* (2023) combines Generative Adversarial Networks (GANs) and Recurrent Neural Networks (RNNs) to highlight the importance of modeling macroeconomic states in predicting individual stock returns. The authors provide evidence that four distinct macroeconomic states effectively capture the dynamics of stock returns.

Both strands have been successfully applied to various tasks, including stock return forecasting, volatility forecasting, asset allocation, and tactical asset allocation. However, because both approaches use time series of asset returns as the main inputs for their models, they are forced to deal with noisy data. This can negatively impact the quality of the regimes uncovered in the data by introducing excessive uncertainty, thus rendering the models less accurate.

This paper extends the tactical asset allocation literature by incorporating macroeconomic regime modeling into the asset allocation process in the form of weighted priors, which effectively addresses some of the issues caused by the noisiness of asset returns data. We extract regime classifications from more stable macroeconomic data rather than asset returns using data-driven clustering techniques from the machine learning literature. We propose a novel model that classifies current regimes, forecasts the distribution of regimes for the next time step, integrates these forecasts with the historical performance of individual assets, and translates the final predictions into portfolio allocations.

Our work utilizes tools from the unsupervised machine learning literature, particularly clustering methods, to derive the regime classification distributions that serve as effective priors for asset allocation. These tools can be broadly divided into algorithmic and model-based approaches. Algorithmic approaches, such as hierarchical, partition-based, and density-based clustering, use iterative processes to classify entities based on similarity measures. A well-known partition-based method is the k-means algorithm (Hartigan and Wong 1979), while DBSCAN is a popular density-based method, effective for discovering clusters of arbitrary shapes and handling noise (Ester *et al.* 1996, Schubert *et al.* 2017).

Our approach is particularly related to fuzzy clustering methods, which extend traditional clustering by allowing data points to belong to multiple clusters with varying degrees of membership. This concept, first introduced by Dunn (1973) and later expanded by Bezdek (1981), provides a more nuanced view of cluster assignments compared to hard clustering methods and allows us to compute classification probability distributions, which are needed to quantify uncertainty, a crucial aspect of integrating regime modeling into asset allocation. In fuzzy clustering, rather than assigning each observation to exactly one cluster, the algorithm assigns membership probabilities across all clusters. This connects naturally with model-based clustering approaches, which rely on probabilistic models, typically using mixture models such as Gaussian mixture models (GMMs), to classify the data, assuming that each cluster follows a distinct probability distribution (Fraley and Raftery 2002).

Our asset allocation model can be broken down into three main stages. The first stage of our model utilizes a novel macroeconomic dataset from the FRED-MD database,

comprising over 100 monthly time series that characterize the state of the US economy. These time series are fed into a modified version of the k-means algorithm. Similarly to the fuzzy c-means model (Bezdek 1981), our algorithm computes probabilities based on centroid distances from any clustering technique. We utilize this method as a wrapper around the k-means algorithm, thus defining the modified k-means algorithm. The modified k-means algorithm then consists of a two-step approach to classify typical and atypical months, and a matching clustering algorithm to ensure consistency of cluster interpretation over time. The output of this stage is a prediction for the current regime and a probability distribution over regimes for the next time step.

In the second stage, we used these regime predictions to build estimates of expected returns and/or volatilities. This can be achieved by using the conditional expected values, given the most likely regimes for the next time step based on past regime occurrences, or by employing a simple ridge regression model estimated using data from the same regime. This process results in forecasts of individual asset returns and/or volatilities.

Finally, the last stage involves mapping these forecasts into asset allocations. We experiment with different configurations of sizing schemes, including equal-weight, long-only, long-and-short, and long with a tactical short tilt when a high-conviction forecast of an impending economic crisis is present.

By pairing this regime classification model with existing portfolio optimization models, we can achieve significantly higher and statistically significant returns, even with relatively simple portfolio optimization models. This empirically demonstrates the value of modeling macroeconomic regime shifts and incorporating that model into a larger investment framework.

Main Contributions. Our work makes several key contributions to the literature on tactical asset allocation and regime modeling. First, we evaluate different methods for computing regime (cluster) uncertainty and demonstrate that a modified k-means algorithm, similar to fuzzy c-means approaches, is better suited for applications requiring smooth, nuanced regime probability transitions over time and overcomes the constraints of deterministic methods. Second, we introduce the use of the comprehensive FRED-MD macroeconomic dataset in a data-driven regime detection model for tactical asset allocation, moving beyond market data alone used by most existing methods and incorporating broader external economic signals. Third, we demonstrate that regime-based portfolios significantly outperform those constructed using random regime classifications. Finally, we demonstrate that our method generates valuable signals, leading to superior performance compared to equal-weight, buy-and-hold, and mean-variance optimization strategies.

2. Related work

Our work relates to and contributes to several strands of literature, particularly in the areas of regime modeling, both from an econometric and machine learning perspective, as well as asset allocation.

Practitioners have extensively used the concept of regime modeling to capture the complexities of financial markets. Regimes are intuitive because they reflect the different phases or states of an economic or financial process, such as periods of high and low volatility, or other stages of the business cycle. Ang and Timmermann (2012) demonstrated that models incorporating regime shifts can capture critical stylized facts about financial time series, including fat tails, time-varying correlations, skewness, and nonlinearities.

Regime modeling in econometrics has a long tradition, with the seminal work by Hamilton (1989) laying the foundation for Markov-switching models. These models have been widely used to analyze many economic and financial processes. Regime-switching models can be applied to stock market prediction modeling, as demonstrated in the works of Schwert (1989), Hassan and Nath (2005), Gupta and Dhingra (2012), and Nguyen (2018). In addition, regime-switching models have been applied to volatility forecasting, with significant contributions from Hamilton and Susmel (1994) and Haas *et al.* (2004). Moreover, these models have been used in monetary policy analysis, as explored by Sims and Zha (2006).

In machine learning, the literature has similarly focused on using asset returns as inputs for regime models. Clustering techniques, such as k-means, have been employed to identify regimes within asset prices for various purposes. For example, Golosnoy *et al.* (2014), Andrada-Félix *et al.* (2016), and Cartea *et al.* (2023) have used various forms of similarity learning or clustering for volatility forecasting. Miori and Cucuringu (2022) and Bennett *et al.* (2022) have applied clustering to lead-lag analysis, while Shu *et al.* (2024) have explored its application for asset allocation using the mean-variance optimization framework.

The asset allocation literature primarily concerns optimizing portfolio returns while managing risk. Regime changes have a significant impact on asset allocation decisions by altering the risk-return profiles of assets. Ang and Bekaert (2004) demonstrated how regime changes can influence asset allocation strategies. Guidolin and Timmermann (2007) provided a comprehensive asset allocation analysis under multivariate regime switching models. Tu (2010) argued that ignoring regime changes can be costly, potentially reducing portfolio returns by 2% per year. The tactical asset allocation literature, closer to what we propose here, focuses on tactical tilts in portfolios to enhance their robustness against adverse periods and minimize drawdowns. This differs from the classical asset allocation literature, which emphasizes long-term strategic asset allocation. Kritzman *et al.* (2012) discussed the implications of regime shifts for tactical asset allocation by modeling regimes with a Markov-switching model.

Most existing methods for incorporating regime shifts into asset allocation rely on detecting distributional shifts in the returns data. Although effective in detecting change points, these approaches are highly sensitive to the choice of market data and do not systematically define regimes based on fundamental market factors. In this work, we bridge the gap between macroeconomic and market data to construct a regime model that integrates external economic signals. This design ensures the applicability of our model across

asset allocation frameworks while providing regimes that are interpretable in terms of fundamental macroeconomic drivers.

Our work also utilizes tools from the literature on unsupervised machine learning and clustering methods. Clustering methods are commonly used to identify underlying patterns or regimes within data, and the literature on clustering can be broadly divided into algorithmic and model-based approaches.

Algorithmic approaches are based on iterative processes that classify entities based on similarity measures and are deterministic in nature. These methods do not provide probabilities for regime membership; instead, they assign entities to specific clusters. Common techniques within this category include hierarchical clustering, partition-based algorithms, and density-based algorithms. For example, the k-means algorithm (Hartigan and Wong 1979) is a widely used partition-based method that minimizes within-cluster variance through an iterative process. At the same time, DBSCAN is a popular density-based method that excels at identifying clusters of arbitrary shapes and handling noise (Ester *et al.* 1996, Schubert *et al.* 2017).

Fuzzy clustering methods, particularly the fuzzy c-means algorithm introduced by Dunn (1973) and refined by Bezdek (1981), bridge the gap between deterministic and probabilistic approaches. Unlike traditional clustering methods that assign each observation to exactly one cluster, fuzzy c-means allows observations to belong to multiple clusters with varying degrees of membership. This is achieved through an optimization problem that minimizes the weighted sum of the squared distances between points and cluster centers, where the weights represent membership degrees. The resulting membership values provide a natural measure of uncertainty in cluster assignments.

In contrast, model-based clustering relies on probabilistic models that assume each cluster follows a distinct probability distribution, enabling the estimation of the probability of each entity's membership across clusters. This probabilistic nature makes model-based approaches, such as GMMs (Fraley and Raftery 2002), particularly suitable for regime identification in financial markets, where capturing uncertainty in cluster assignments is valuable.

3. Methodology

We introduce a data-driven regime detection method that defines monthly regime distributions based on macroeconomic variables. Our approach begins with a regime classification strategy that leverages k-means clustering with distinct distance functions (Section 3.1). We then extend this framework by relaxing the deterministic nature of k-means to produce probabilistic regime assignments (Section 3.2). Finally, we show how to derive the regime transition probability matrix within this framework (Section 3.3).

3.1. Regime classification

We utilize the k-means algorithm with various distance functions to generalize existing strategies employed by financial

analysts for categorizing previous economic periods. We start by selecting a large set of m macroeconomic variables that represent the market's broader economic state for each month. For each month t being categorized, we have a vector $x_t \in \mathbb{R}^m$ that captures the market state at time t .

3.1.1. Separating outliers using ℓ_2 clustering. The first partition we make of our data set is to label and separate outlier months. The economic intuition behind this choice of the first step is to identify months with extraordinary market conditions that may not accurately represent market dynamics in the rest of the dataset. We use the ℓ_2 distance between each pair of months to perform k-means clustering on our dataset with $k = 2$. Because the ℓ_2 distance is *sensitive to outliers*, this process effectively prunes the deviant months and leaves behind the most typical months.

After labeling each month, we are left with two groups: A and B . Without loss of generality, we assume that A and B are such that $|A| \leq |B|$. In this case, B can be considered the cluster of *typical months* and A the cluster of *outliers*. We begin our regime classification by labeling the months in A as REGIME 0, and turn our attention to the months in B for the remainder of the regime classification algorithm.

3.1.2. Splitting typical samples using cosine clustering.

After removing outliers, the months in the set B can be considered periods representing ‘business as usual.’ The ℓ_2 clustering used to define the set B ensures that the state vectors of the months within B have magnitudes of similar sizes. To further parse these periods requires switching to a distance metric that is *magnitude agnostic*. Cosine similarity uses the cosine of the angle between two vectors as a measure of distance. It ignores the magnitude of those vectors in its computation, rendering it an apt choice. Therefore, we compute the cosine similarity for each pair of months in B , then use a k-means clustering algorithm with $k = r$ to arrive at the final regime classifications for every month in our dataset. We use the simple k-means ‘elbow’ heuristic to determine the optimal value of r during cosine clustering.

After both rounds of clustering, we have classified each month into one of the $r + 1$ regimes. REGIME 0 represents outlier months with abnormal macroeconomic conditions, and the remaining regimes in $\{\text{REGIME } 1, \dots, \text{REGIME } r\}$ represent different types of regular market behavior.

The Algorithm 1 provides an overview of the regime classification process. This algorithm returns probability distributions over regimes rather than discrete regimes using a mechanism described in Section 3.2. These regime distributions can then be used to improve forecasting models or decision criteria through probabilistic conditioning.

3.2. Computing regime probability distributions

While the process described so far is intuitive and can produce sufficient regime classifications for every month, to make this tool useful in practice, it should be able to output *probability distributions* for each regime instead of just discrete classifications. This computation is not trivial in our case due to two

Algorithm 1 An overview of the regime classification process

Require: $X = \parallel_{t=1}^T x_t$ ($\parallel$ denotes the concatenation operation) is the dataset consisting of the concatenation of all monthly macroeconomic state vectors x_t for $t \in [1, T]$.

```

{A, B} ← KMEANS[ℓ2](X, 2)
if |A| ≤ |B| then
  R0 ← A
  r ← KMEANSELBOWVALUE[COSINE](B)
  {R1, ..., Rr} ← KMEANS[COSINE](B, r)
else
  R0 ← B
  r ← KMEANSELBOWVALUE[COSINE](A)
  {R1, ..., Rr} ← KMEANS[COSINE](A, r)
end if
return {R0, ..., Rr}

```

main factors. To start, the k-means algorithm does not have a direct stochastic algorithm implementation in the sense of cluster labels. Secondly, two different versions of the k-means algorithm are used consecutively on various subsets of X , so any probabilistic output must be a combination of the outputs of both instances of the k-means algorithm.

3.2.1. Probabilistic distributions from the K-Means algorithm.

We address the first issue by leveraging the fact that, for any given sample to be classified, the k-means algorithm utilizes the distance between that sample and the cluster centroids of each cluster to classify it. The distances can, therefore, be interpreted as measures of how ‘close’ that sample is to being classified as each respective cluster. For a given vector x whose distance from centroid i is d_i , we define the probability of that vector being in cluster C_i as

$$P(C_i) = \frac{1 - \frac{d_i}{\sum_j (d_j)}}{\sum_m \left(1 - \frac{d_m}{\sum_j (d_j)}\right)} \quad (1)$$

We emphasize that this procedure is essentially the same as the fuzzy c-mean, although implemented in a slightly different fashion. We use this method to compute the distributions for KMEANS_[ℓ₂] and KMEANS_[COSINE].

3.2.2. Combining probabilities from KMEANS_[ℓ₂] with those from KMEANS_[COSINE].

Combining the distributions from KMEANS_[ℓ₂] and KMEANS_[COSINE] to get a full distribution over all regimes is not as simple as multiplying the distribution from KMEANS_[COSINE] by $1 - P(\text{REGIME } 0)$ because doing this can diffuse $1 - P(\text{REGIME } 0)$ too thinly, causing the resulting distribution to be inconsistent with the output of Algorithm 1. Instead, we compute the probability distribution of $\{\text{REGIME } 1, \dots, \text{REGIME } r\}$ using the previously described method and then separately calculate a new *scaled value* P_{R0} from $P(\text{REGIME } 0)$. To this end, we first define

$$P_{\max} = \max(\{P(\text{REGIME } 1), \dots, P(\text{REGIME } r)\}). \quad (2)$$

We then choose the value for P_{R0} by recognizing that the probability distribution for $\text{KMEANS}_{[\ell_2]}$ can be thought of as a measure of how close P_{R0} should be to $P_{\max}$. Before normalizing the distribution to sum to 1, the following must therefore hold

$$P_{R0} = \begin{cases} 0 & P(\text{REGIME } 0) = 0.0 \\ P_{\max} & P(\text{REGIME } 0) = 0.5 \\ \infty & P(\text{REGIME } 0) = 1.0 \end{cases} \quad (3)$$

One such continuous function that would satisfy these conditions cleanly is

$$P_{R0} = -P_{\max} \log_2(1 - P(\text{REGIME } 0)). \quad (4)$$

We use equation (4) to set P_{R0} , then renormalize the distribution to obtain the final regime probability distribution.

3.3. Computing the regime transition probability matrix

After classifying each month into a regime, computing the transition probability matrix between regimes sheds light on how market dynamics change monthly and describes the relations between regimes.

Each matrix entry can be computed by counting the number of times a given regime switches to every other regime and dividing that by the number of times that regime appears in the dataset. To compute an element e_{ij} of the regime transition probability matrix, we write

$$e_{ij} = \frac{\text{TRANSITIONCOUNT}(\text{REGIME } i, \text{REGIME } j)}{|\text{REGIME } i|}. \quad (5)$$

Figure 5 shows an example of such a transition probability matrix.

4. Regime analysis

Next, we present details of the macroeconomic data we are using (Sections 4.1-4.2), compare methods for computing the financial regime associated to each month within the dataset (Section 4.3), and present an analysis of those regimes (Section 4.4) along with what they reveal about the corresponding market behavior (Sections 4.5-4.6).

4.1. Macroeconomic data

In this work, we used 127 variables from the FRED-MD dataset to represent the macroeconomic state of the market. The FRED-MD dataset is a monthly public database of macroeconomic data collected by the Federal Reserve Bank of St. Louis, created by McCracken and Ng (2016) to make data-oriented economic research more accessible. FRED-MD is composed of a variety of groups of variables, namely

- (i) Output and Income
- (ii) Consumption, Orders, and Inventories
- (iii) Labor Market

- (iv) Housing
- (v) Money and Credit
- (vi) Interest and Exchange Rates
- (vii) Prices

We excluded group (vi) to focus only on US macroeconomic data.

The variables are sampled monthly and organized into a data frame indexed by timestamps representing the first day of each month. Each row of this data frame then represents the macroeconomic state of the market for that month. This work uses macroeconomic data from December 1959 to January 2023.

4.2. Data preparation

The FRED-MD variables must be standardized before the dataset can be analyzed. To properly perform this standardization, most variables in the FRED-MD dataset must be transformed before they can be demeaned and standardized to account for their different units (indices, prices, rates, etc.). The appropriate transformations to apply before standardization are pre-defined by the Federal Bank of St. Louis (each variable has a *t-code* assigned to it that corresponds to one of seven possible transformations).

We apply these transformations, standardize the variables, and then use principal component analysis (PCA) to find the eigenvectors of the data with the largest corresponding eigenvalues. These are used to reduce the number of dimensions in the dataset and extract the most critical factors. The amount of variance explained by each component can be cumulatively summed in descending order of their eigenvalues until a pre-set variance explanation threshold is met. In this work, we use a threshold of 95%, corresponding to 61 components. A visualization of the variance explained by varying the number of components is shown in figure 1.

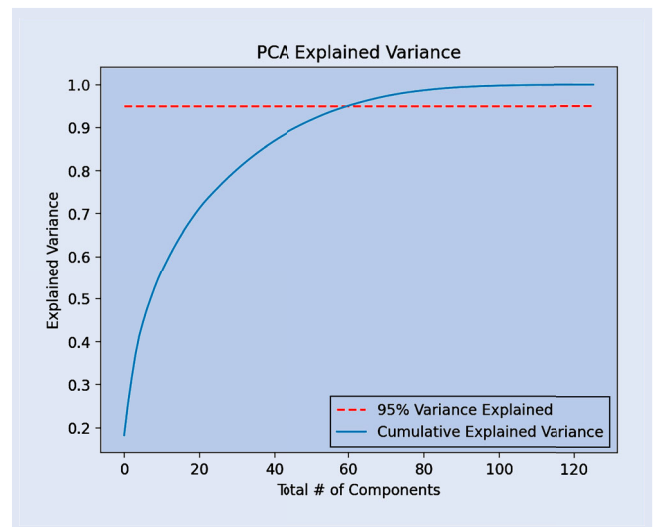


Figure 1. The cumulative variance explained from using different amounts of eigenvector components in the PCA dimensionality reduction process. The curve reaches 95% at around the 61-component mark.

We also set the k value of our algorithm's second k-means clustering layer using the previously mentioned k-means 'elbow' heuristic. The optimal value of k to use in the $\text{KMEANS}_{[\text{COSINE}]}$ part of our methodology is $k=5$. We therefore set k to this value in this work.

4.3. Clustering method comparison

This section compares the modified k-means method with the Gaussian Mixture Model (GMM) to illustrate the differences between an algorithmic clustering approach and a model-based one. Although the GMM is inherently probabilistic and efficiently extracts probabilities for each regime assignment, the modified k-means relies on a heuristic to estimate probabilities, offering a different perspective on cluster confidence and transitions.

Figure 2 compares the regime labels assigned by the modified k-means and GMM methods. The gray-shaded areas denote NBER-defined recession periods. Both methods capture similar regimes, aligning on most classifications, with only minor differences observed in transitions, particularly for regimes 2 and 3. This suggests that while both methods are robust in identifying consistent patterns, their unique sensitivities to data structure lead to slight variations. Both methods also strongly align with the NBER-defined recession periods, confirming their ability to detect macroeconomic shifts effectively.

Figure 3 compares crisis probabilities derived from the modified k-means heuristic and the GMM. While the k-means probabilities exhibit smoother transitions over time, suggesting a more nuanced approach to modeling uncertainty,

the GMM assigns almost binary probabilities that reflect strong confidence in its cluster assignments. This probabilistic framework of the GMM ensures clarity in classification but may overlook subtle transitions. On the other hand, the gradual shifts in the k-means probabilities highlight its flexibility in capturing heterogeneity within clusters, where it adjusts probabilities to reflect varying levels of certainty as new data becomes available.

Interestingly, both methods align on specific probability spikes, indicating agreement on key transition points, particularly during significant economic turning points. This alignment underscores the reliability of these clustering approaches in identifying critical periods of economic shifts. However, the modified k-means method offers a more interpretable view of uncertainty, rendering it particularly valuable for applications such as portfolio allocation, where understanding gradual transitions rather than abrupt changes can guide better risk management and decision-making strategies.

Overall, this comparison highlights the trade-offs between the two approaches. The GMM provides high-confidence regime classifications that are nearly deterministic, making it suitable for applications requiring clear, binary outcomes. In contrast, the k-means approach strikes a balance between flexibility and interpretability, capturing the gradual dynamics of regime transitions. Both methods exhibit strong consistency with NBER-defined recession periods, demonstrating their ability to effectively detect macroeconomic shifts. However, the probabilistic nuance of the k-means method provides an additional layer of insight into regime dynamics, which can be critical for understanding the complexities of financial markets.

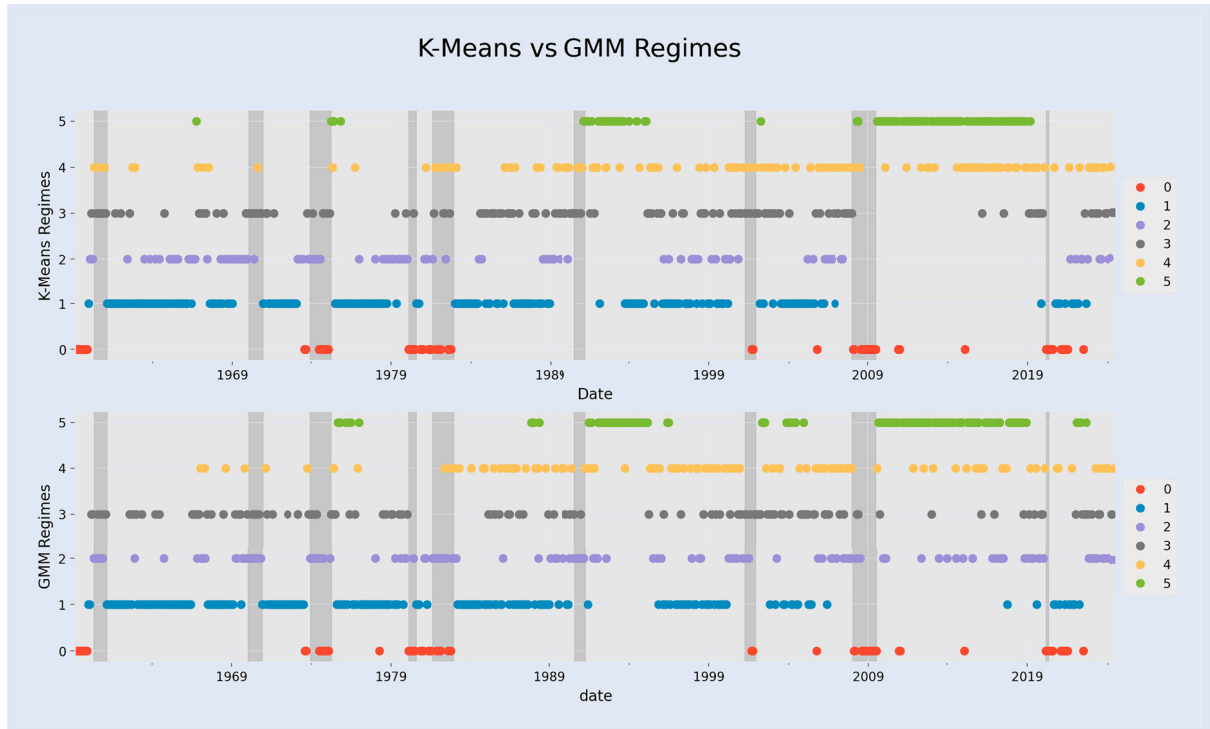


Figure 2. A comparison of the regimes detected using the layered k-means method and the analogous GMM method. NBER recession periods are highlighted, alongside regime labels for both methods. Most classifications align, with notable differences between the two methods observed in REGIME 2 and REGIME 3.

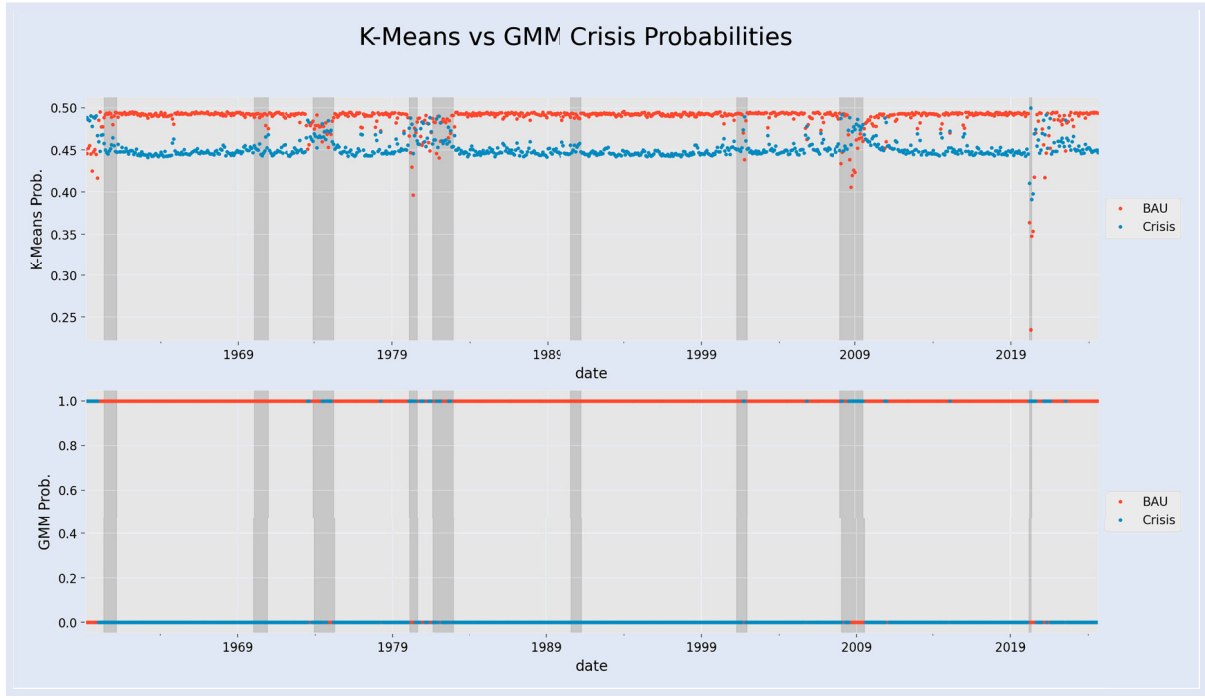


Figure 3. A comparison of the probabilities of being in a crisis period versus a normal period (labeled ‘BAU’ for ‘business as usual’), as computed using the k-means distribution heuristic and the GMM. NBER recession periods are highlighted. The k-means probabilities exhibit gradual transitions, whereas the GMM distributions appear deterministic.

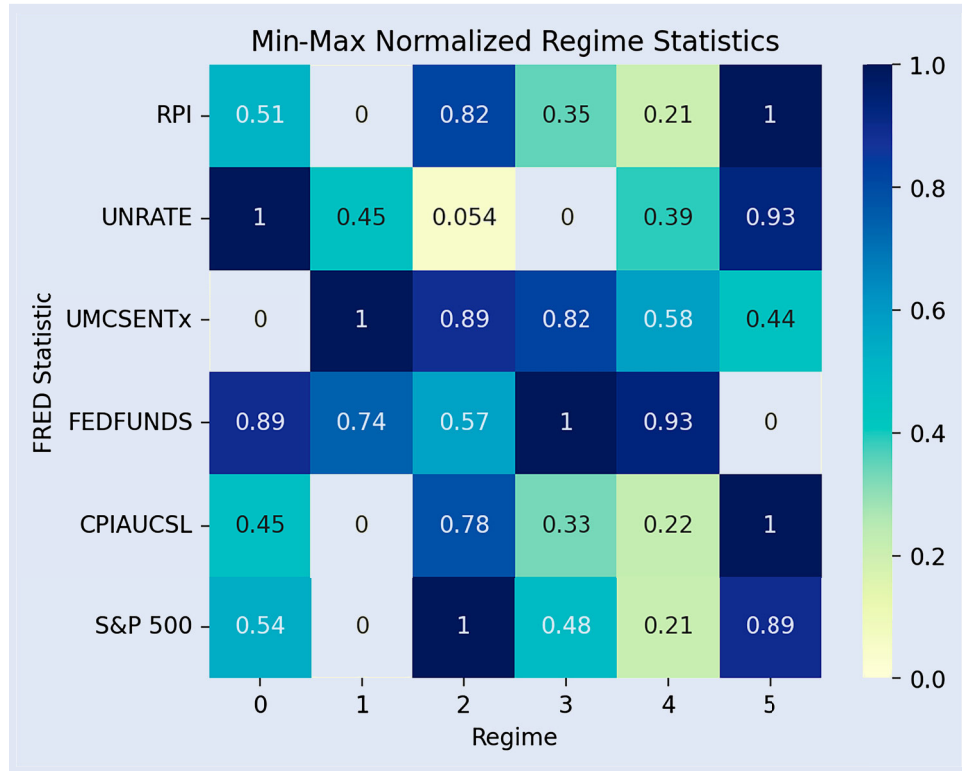


Figure 4. Macroeconomic differences across regimes as measured via the average value of crucial FRED-MD statistics. The value of the statistics is min-max normalized along each row to make them easier to compare with one another.

4.4. Regime properties

One can understand the economic differences between regimes by measuring the average value of certain crucial FRED-MD statistics within each regime. Figure 4 provides this type of analysis.

Here, we observe that REGIME 0, the regime containing all of the ‘outliers’ months filtered out using the $KMEANS_{[\ell_2]}$ algorithm, best corresponds to periods of economic difficulty. To support this intuition, REGIME 0 has the lowest real personal income, consumer sentiment, federal funds rate, and the highest unemployment rate. This implies that the effect of

stacking $\text{KMEANS}_{[\ell_2]}$ and $\text{KMEANS}_{[\text{COSINE}]}$ in the regime labeling process is first to separate the atypical periods of economic panic, then cluster the remaining months of normal economic activity based on the direction of their macroeconomic state vectors.

Because the clustering is performed on PCA factors derived from all 127 series, the statistics shown here are interpretability summaries computed *after* clustering by averaging selected representative indicators within each regime.

4.5. Market behavioral analysis using regimes

Using the regime transition probability matrix computed from the monthly regime timeline, we can now understand how

regimes are related and how a market is likely to change over time. We present such a matrix in figure 5, and draw the following conclusions.

We observe that most of the probability density lies on the diagonal. This indicates that the most likely regime for the next month is always the current month's regime, i.e. regime shifts do not occur with high frequency, which is what we expect to observe based on the existing literature on macroeconomic regimes, such as Baele *et al.* (2015).

We also note that some matrix elements have shallow values, some being precisely 0. This implies that specific regime transitions are either impossible or extremely unlikely, which is also what we would expect to see in a well-functioning macroeconomic regime detection model.

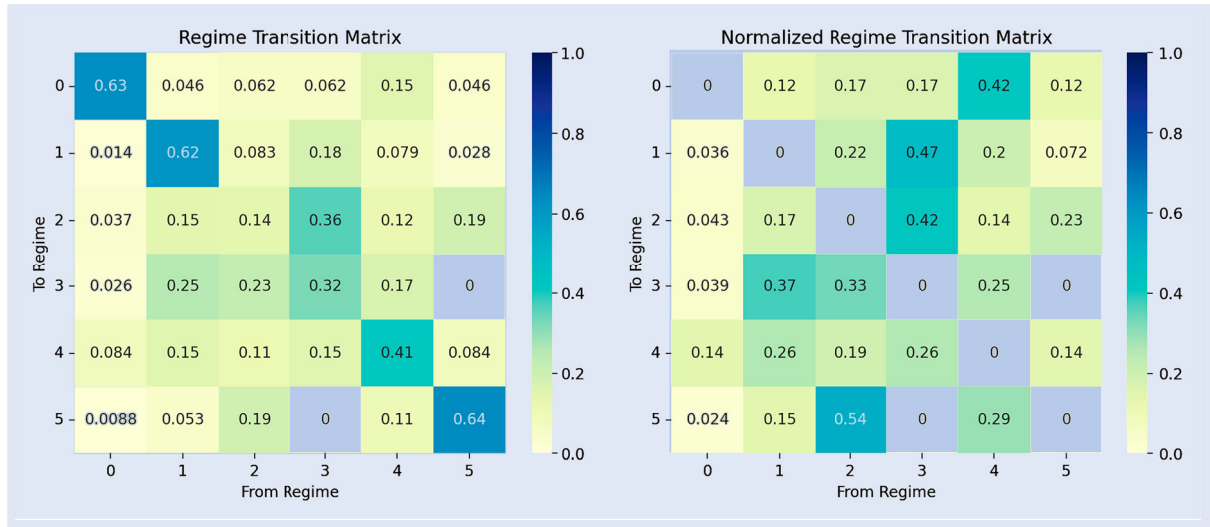


Figure 5. *Left*: Regime transition probability matrix computed using all available data. *Right*: Regime transition probability matrix given that a regime transition occurs.

Table 1. Economic characterizations of each regime.

Regime index	Regime label	Characteristic description
0	Economic Difficulty	This regime has the highest unemployment rate and lowest consumer sentiment, federal funds rate, and real personal income. This regime is also the most self-stable and likely to transition to stagflation.
1	Economic Recovery	This regime represents a recovery phase where inflation is controlled, but equity markets lag due to uncertainty. High consumer sentiment here suggests there is public confidence in future growth.
2	Expansionary Growth	This regime has strong economic prosperity. Good equity market performance and high consumer sentiment reflect robust economic confidence. Inflation is moderate, and unemployment is at a balanced level, creating favorable conditions for growth. Central banks typically maintain a neutral monetary policy.
3	Stagflationary Pressure	This regime is marked by high inflation and rising interest rates. While unemployment may not be extreme, the restrictive monetary policy dampens equity markets. This regime aligns with stagflationary environments where growth stagnates alongside rising prices.
4	Pre-Recession Transition	This regime could be a warning sign before a recession. Inflation begins to cool, but economic activity slows, leading to subdued equity markets. Unemployment rises slightly, and monetary policy begins to tighten in an effort to combat earlier inflation, which can trigger an economic downturn.
5	Reflationary Boom	This regime reflects periods of economic resurgence fueled by loose monetary policy and strong equity market performance. Inflation is high, but central banks prioritize growth and employment over price stability. These periods often coincide with quantitative easing or other accommodative measures.

Note: Regimes are learned from PCA factors computed from all 127 FRED-MD variables, and descriptions summarize post-hoc within-regime averages of representative indicators.

Link between the 127 FRED–MD variables, regime clustering, and table 1. The regime labels are learned from the full information set of 127 transformed and standardized FRED–MD series (Section 4.2). In particular, we apply PCA to this panel and retain the first 61 components (95% explained variance), and the k-means layers cluster monthly observations in this reduced factor space. Therefore, each regime is *defined* by a multivariate pattern across the entire macroeconomic panel rather than by any single indicator.

To make the regimes economically interpretable, we *post-characterize* them by computing within-regime averages of a small set of representative macroeconomic indicators (e.g. real activity, labor market conditions, prices/inflation, policy rates, sentiment, etc.), which are either directly contained in FRED–MD or closely aligned proxies within the dataset’s variable groups (Output and Income, Labor Market, Prices, etc.). Table 1 summarizes these descriptive patterns and should be read as an interpretation of the regimes implied by the full 127-variable clustering, not as the input definition of the regimes.

Finally, the comparison to NBER recessions in figures 2–3 is included as an external validation: NBER recession dates are determined using a broad set of real activity indicators (e.g. income, employment, production, and sales). Our regimes, learned from a wide macro panel that includes these dimensions, align closely with NBER recession periods, indicating that the clustering recovers business-cycle-like structure from the data.

4.6. Characterizing regimes

We characterize and label each regime descriptively using the information in figures 4 and 5. Although these characterizations cannot be absolute due to the approximate nature of the k-means algorithm, we present our regime descriptions in table 1.

To help us better understand these regimes, we further modify the transition probability matrix in figure 5 to represent transition probabilities *given that a transition takes place*. This can be achieved by dividing all the off-diagonal elements of row i by $1 - \text{DIAG}(i)$, where $\text{DIAG}(i)$ is the i -th diagonal element of the transition probability matrix, then setting all the diagonal elements to 0. This variation of the regular transition probabilities helps us better understand the relationships between *different regimes*. Without this normalization, regimes that tend to last longer would have reduced probabilities of transitioning to other regimes, which would introduce bias into downstream comparisons by making it appear that these regimes are not closely connected with other regimes. Our normalization accounts for this and has the same effect as normalizing our probabilities by the expected duration of any regime. After this step, we arrive at the normalized matrix in figure 5.

Using this new matrix, we now define a graph structure over the regimes using the matrix from figure 5 as the graph adjacency matrix. Here, the edge weights are characterized by the probability that one regime will transition to another, given that a transition occurs.

The graph from this procedure, shown in figure 6, helps to better understand how the market changes over time. The

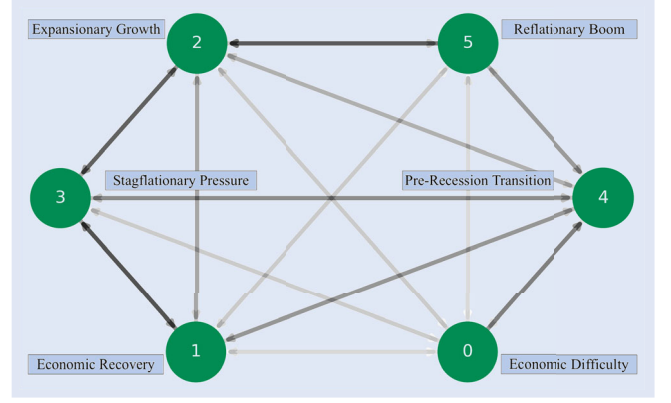


Figure 6. Regime transitions visualized in network form. Darker edges signify larger transition probabilities, and nodes have been positioned to emphasize the emergent path graph between the regimes. Labels are taken from table 1.

edges are colored according to their edge weight, and the positions of the nodes have been rearranged to highlight the strong path graph that emerges from the regime data.

4.7. Temporal validation of regime centroids

The regime classification introduced in Section 3.1 is obtained by applying a two-stage clustering procedure to an expanding-window principal component representation of the FRED–MD panel. In this setting, each regime is characterized not only by its conditional averages in the original macroeconomic variables (figure 4), but also by a sequence of *centroids* that evolve over time in latent principal component space.

Before analyzing the behavior of centroids around specific macroeconomic events, it is useful to clarify their construction and economic role. In the first stage, an ℓ_2 -based KMeans clustering is used to separate atypical observations from the bulk of the data, producing coarse clusters that distinguish extreme or outlier months (e.g. pandemic-related disruptions) from more typical macroeconomic conditions. The resulting ℓ_2 centroids therefore provide a high-level partition of the sample into ‘hard,’ ‘easy,’ and outlier periods, capturing large-scale deviations in the latent macroeconomic state.

In the second stage, cosine-distance KMeans clustering is applied within the set of typical observations to refine the regime structure. This step yields a set of cosine centroids that represent distinct macroeconomic regimes within the typical portion of the sample. Importantly, these centroids are defined in the same principal component space used for clustering and can be tracked over time as the expanding window incorporates new information.

Figure 7 reports the raw time series of the L2 and cosine centroids. The $\text{KMEANS}_{[\ell_2]}$ centroids exhibit sharp movements during well-known episodes of macroeconomic stress, most notably during the COVID-19 period, confirming their role in isolating extreme states of the economy. In contrast, the cosine centroids display richer, higher-frequency variation, reflecting the finer decomposition of typical macroeconomic regimes.

While the raw centroid series already reveal meaningful structure, their high-frequency variability can obscure

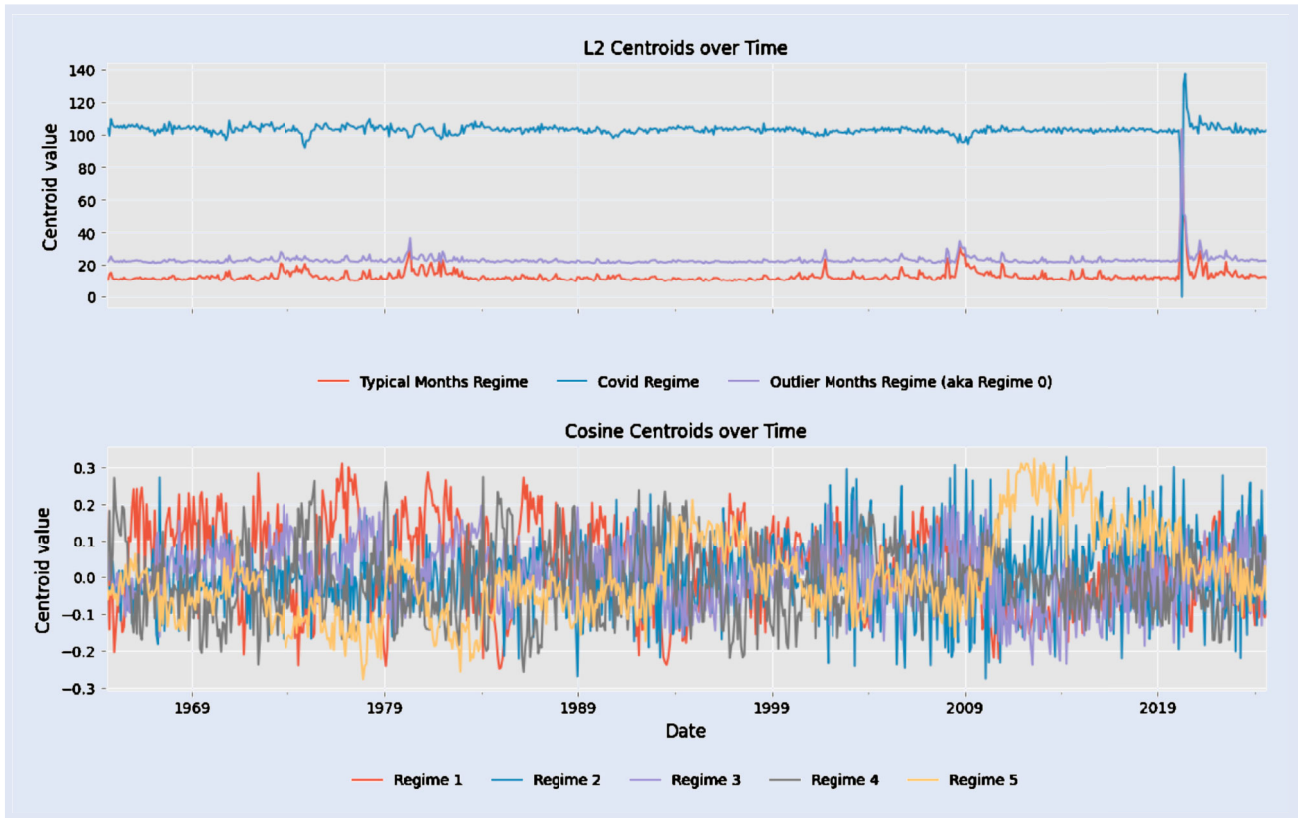


Figure 7. Raw regime centroids over time in principal component space. The top panel reports L2 centroids used to separate outlier and typical observations, while the bottom panel reports cosine centroids corresponding to refined macroeconomic regimes within the typical sample.

longer-run patterns associated with macroeconomic cycles. To facilitate interpretation, we therefore compute rolling six-month moving averages of the centroid time series. This smoothing step does not alter the underlying clustering results, but allows for a clearer visualization of how regime centroids evolve in response to persistent macroeconomic conditions.

Figure 8 presents the six-month moving averages of the $KMEANS_{[COSINE]}$ centroids over the full sample. The smoothed series reveal pronounced medium-term dynamics, with sustained deviations that align with well-documented macroeconomic episodes. This observation motivates a more targeted analysis of centroid behavior around specific events.

Figure 4 characterizes the regimes in terms of conditional averages of economically interpretable macroeconomic variables. While these summaries provide a static description of each regime, the clustering procedure itself is conducted in a lower-dimensional principal component representation of the FRED-MD panel. As a result, an important interpretability question remains: do the regime centroids inferred in principal component space evolve over time in a manner that is consistent with observed macroeconomic conditions?

To address this question, we analyze the temporal dynamics of the regime centroids by computing rolling six-month moving averages of the centroid series obtained under the expanding-window clustering procedure. Recall that, at each date, centroids represent the mean location of the optimal

clusters in principal component space. If the inferred regimes capture economically meaningful states of the macroeconomy, then these centroids should respond in a systematic and intuitive way to major macroeconomic events reflected in the underlying FRED-MD variables.

We focus on two classes of events that are particularly informative in this regard: periods of elevated inflation and officially dated U.S. recessions. In each case, we examine the behavior of the centroid moving averages in windows surrounding these events. The regimes highlighted in the analysis correspond to those exhibiting the highest values in figure 4, which identifies the regimes most strongly associated with high inflation and monetary tightening conditions.

High-Inflation Episodes. Figure 9 reports the six-month moving averages of the regime centroids around several historically prominent inflationary episodes, including the Great Inflation era (1960–1980), the post-Volcker stabilization period (1985–1995), the late-1990s expansion, and the post-COVID inflation surge. In each panel, the highlighted regime centroids display pronounced and persistent deviations during inflationary phases, while other regimes remain comparatively muted. Notably, the magnitude and persistence of these movements are largest during episodes commonly associated with sustained inflationary pressure, such as the 1970s and the post-2020 period.

This behavior is consistent with the economic interpretation suggested by figure 4: regimes characterized by high inflation and restrictive monetary policy exhibit centroid dynamics

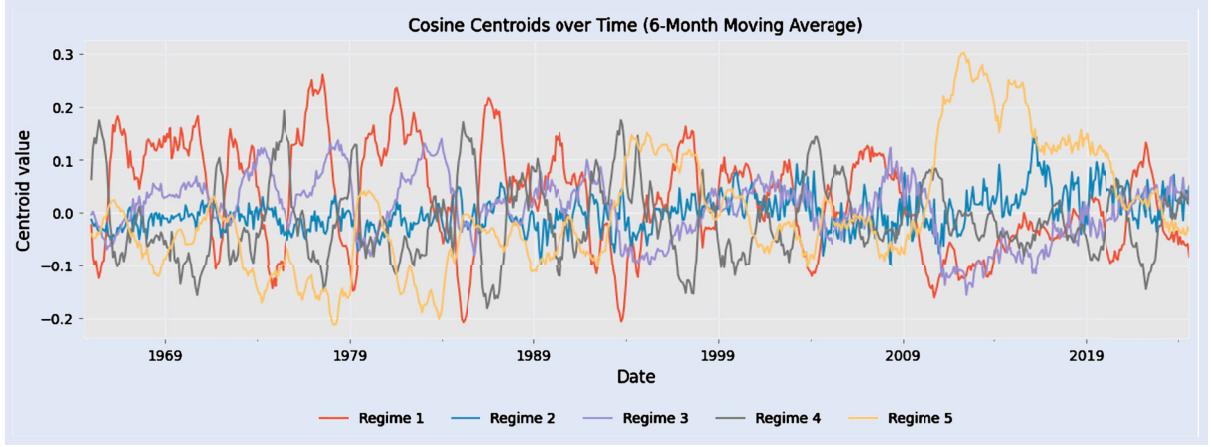


Figure 8. Six-month moving averages of cosine regime centroids over the full sample. Smoothing highlights medium-term regime dynamics in principal component space.

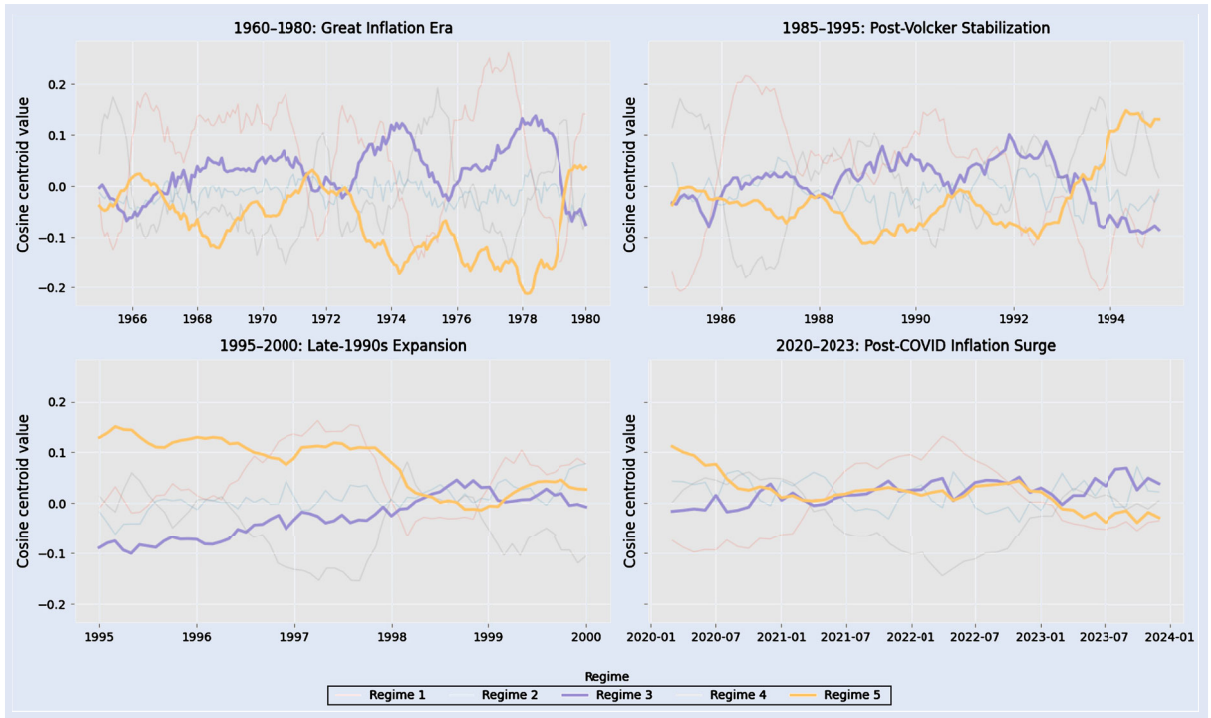


Figure 9. Six-month moving averages of regime centroids around major U.S. inflationary episodes. Highlighted lines correspond to regimes most strongly associated with high inflation and monetary tightening, as identified in figure 4.

that intensify during inflationary cycles and attenuate during periods of price stability.

Recessions. Figure 10 presents an analogous analysis centered on NBER-dated U.S. recessions, including the 1973–75, 1981–82, 2001, and 2007–09 downturns. The shaded regions indicate official recession periods. Across all episodes, the centroid dynamics of the highlighted regimes exhibit systematic changes during recessions, with clear directional shifts that coincide with deteriorating macroeconomic conditions. In contrast, regimes associated with expansionary or recovery phases tend to weaken or reverse during these intervals.

These patterns reinforce the interpretation that the inferred regimes correspond to coherent economic states rather than artifacts of high-dimensional clustering. In particular, the fact that centroid movements align with well-documented macroeconomic turning points provides additional evidence that the

principal-component-based clustering captures economically relevant structure in the data.

5. Forecasting experiments

We now demonstrate how these regimes can be leveraged to enhance forecasting models and decision criteria, thus improving the performance of portfolios based on their outputs.

For this work, we chose to focus on a tactical asset allocation task for a portfolio composed solely of ETFs and index funds, as we anticipate that these are the types of assets most likely to be affected by the market's macroeconomic conditions.

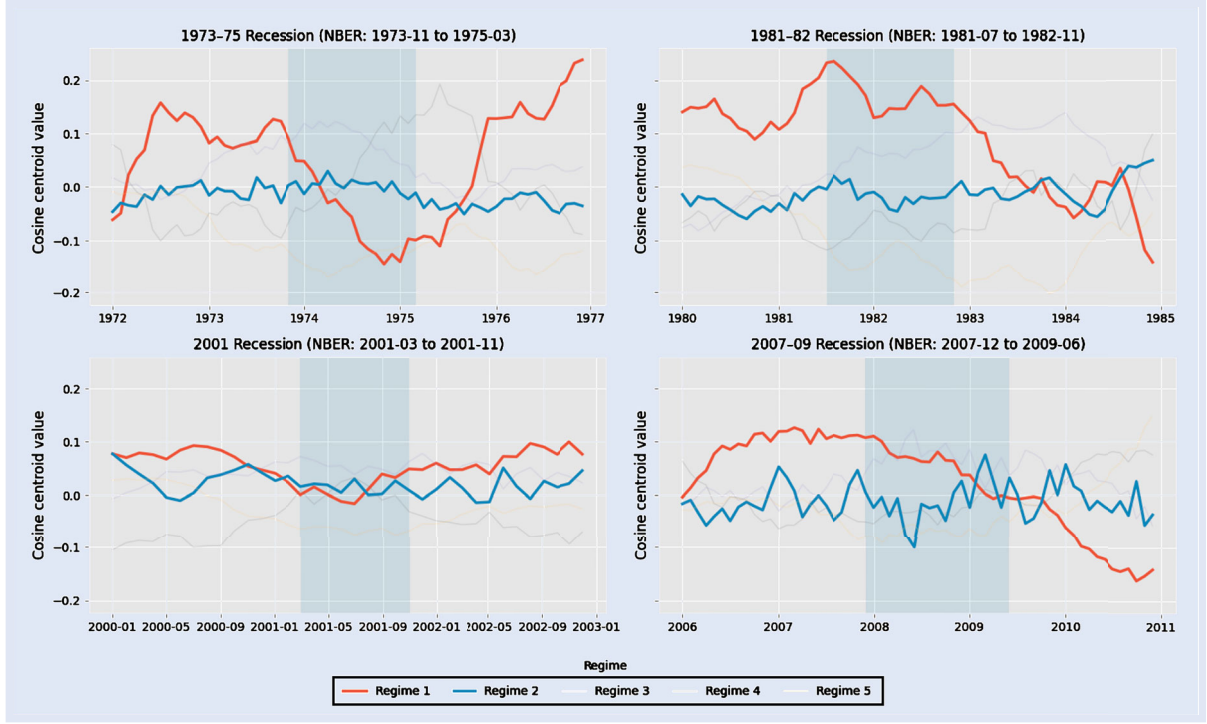


Figure 10. Six-month moving averages of regime centroids around NBER-dated U.S. recessions. Shaded areas denote official recession periods. Highlighted regimes correspond to those most closely associated with contractionary macroeconomic conditions.

5.1. Regimes as probabilistic conditions

Regimes can improve forecasting models and decision criteria by serving as a probabilistic condition on the forecast of the underlying asset. Ideally, we should consider the current regime estimate and the regime transition probability matrix, which represents the most probable following regimes. It is essential to acknowledge the uncertainty surrounding these estimates. When dealing with the uncertainty around current regimes, instead of treating the output label for the current month as a deterministic fact, we should consider the distribution of possible current regimes.

In practice, this means combining the distribution over the current regimes and the regime transition probability matrix before applying it to any chosen forecasting model or decision criteria. To describe this procedure more precisely, recall that $P(\text{REGIME } i)$ is the probability measure associated with the regimes i , for $i = 0, \dots, r$, induced by the procedure described in Section 3.2. Let $\mathbf{p}_t \in \mathbb{R}_+^{r+1}$ be the discrete probability values associated with each regime for the t -th time step, and let $E_t \in \mathbb{R}_+^{(r+1) \times (r+1)}$ denote the regime transition probability matrix defined in Section 3.3.

As mentioned at the end of Section 3.2, we first need to ensure that $\mathbf{p}_t$ is renormalized to sum to 1. We achieve this by dividing it by its magnitude as follows

$$\tilde{\mathbf{p}}_t = \frac{\mathbf{p}_t}{|\mathbf{p}_t|}. \quad (6)$$

Second, we update the current state probability by combining both the distribution over current regimes and the transition matrix

$$\tilde{\mathbf{p}}_{t+1} = \tilde{\mathbf{p}}_t^\top E_t. \quad (7)$$

In fact, this process is identical to that of iterating a Markov chain process (Ross 2019). It is important to note that, as a less effective alternative, we could fix the current state to be the one with the highest probability. However, this process would completely ignore the uncertainty surrounding the current state estimates and would result in a model whose probabilistic conditioning would remain identical for months in the same regime, despite the macroeconomic state vectors being different.

5.2. Forecasting models

We employ two simple forecasting strategies and two decision criteria models to illustrate how knowledge of current market conditions and transition probabilities can enhance model accuracy. The forecasting models used are the Naive Forecasting Model and the *Linear Ridge Regression Model*, whereas the decision criteria models are the *Mean-Variance Optimization Model* and the *Black-Litterman Model*.

The Naive Forecasting Model generates predictions using past Sharpe ratios conditioned on the most probable regime. The Linear Ridge Regression Model constructs specialized models by estimating parameters for each market regime and uses these estimates to make forecasts. Final predictions are adjusted based on updated estimates of current regime probabilities, as defined in equation (7). The Black-Litterman Model follows the framework proposed by Black and Litterman (1991), combining the sample mean and covariance of returns with priors derived from our proposed regime model. Finally, the Mean-Variance Optimization Model incorporates no regime information and serves as a benchmark.

Let $\hat{y}_{t+1}^j$ represent the prediction for the time step $t+1$ made by any of the models considered for the ETF j . Let $s_{1:t}^j$

denote the Sharpe ratio computed using the returns from 1 to t . Lastly, let $\mathbf{X}_{1:t}^i$ be a matrix of macroeconomic characteristics observed in REGIME i from 1 to t . Next, we will define each of the forecast models using this notation.

5.2.1. Naive forecasting model. The naive forecasting model is a simple forecasting scheme designed to evaluate the effectiveness of the regime module. It is parameter-free and relies solely on historical data conditional on regime classifications.

For each time step $t+1$, we first identify the most likely regime i_{t+1}^* , defined as

$$i_{t+1}^* = \underset{i \in \{1, \dots, r+1\}}{\operatorname{argmax}} \tilde{p}_{i,t+1}. \quad (8)$$

Given this predicted regime, the forecast for ETF j at time step $t+1$ is the expected Sharpe ratio conditional on the regime

$$\hat{y}_{t+1}^{j,\text{naive}} = \mathbb{E}[s_{j,t+1} | \text{REGIME} = i_{t+1}^*] = \hat{s}_{j,1:t}^*, \quad (9)$$

where $\hat{s}_{j,1:t}^*$ is the sample estimate of the conditional Sharpe ratio

$$\hat{s}_{j,1:t}^* = \frac{\hat{\mu}_{j,1:t}^*}{\hat{\sigma}_{j,1:t}^*} \quad (10)$$

Here, $\hat{\mu}_{j,1:t}^*$ and $\hat{\sigma}_{j,1:t}^*$ are the sample mean and standard deviation, respectively, calculated using data from time steps 1 to t where the regime matches i_{t+1}^* . Thus, the estimation in equation (9) uses only the subset of historical data corresponding to the predicted regime.

5.2.2. Linear ridge regression model. The linear ridge model proposed by Hoerl and Kennard (1968, 1970) is a step further using simple learning models to predict the next time step returns. There are two main differences between this model and the naive benchmark: (1) it predicts returns instead of Sharpe ratios, and (2) it has learnable parameters that get updated with each new time step.

Following up the intuition built above, the forecast of the ridge model for each REGIME i and ETF j is given by

$$\hat{y}_{t+1}^{i,j,\text{ridge}} = \mathbb{E}[r_{t+1}^j | \text{REGIME } i] = \hat{\beta}_{\text{ridge}}^{i,j} \mathbf{X}_{1:t}^i, \quad (11)$$

where

$$\begin{aligned} \hat{\beta}_{\text{ridge}}^{i,j} = \underset{\beta^{i,j}}{\operatorname{argmin}} & \left\{ (\mathbf{y}_{1:t}^j - \mathbf{X}_{1:t}^i \beta^{i,j})^\top (\mathbf{y}_{1:t}^j - \mathbf{X}_{1:t}^i \beta^{i,j}) \right. \\ & \left. + \lambda \|\beta^{i,j}\|_2^2 \right\}. \end{aligned} \quad (12)$$

Once we have available all the forecasts for each ETF and the regime, we again use the values in $\tilde{\mathbf{p}}_{t+1}$ to aggregate the forecasts through the regime dimension such that we end up with a final probability weighted forecasts for each ETF, as follows

$$\hat{y}_{t+1}^{j,\text{ridge}} = \sum_{i=1}^r \tilde{p}_{i,t+1} \cdot \hat{y}_{t+1}^{i,j,\text{ridge}}. \quad (13)$$

5.3. Portfolio decision criteria

5.3.1. Standard decision criteria. Let $\hat{\mathbf{y}}_{t+1}^k \in \mathbb{R}^d$, for $k \in \{\text{naive}, \text{ridge}\}$ be a vector of predictions for all d ETFs. Let also $\mathcal{H}$ and $\mathcal{L}$ be two sets composed of the l highest and l lowest values of $\hat{\mathbf{y}}_{t+1}^k$ respectively. We assume that $\mathcal{H}$ has only positive values and scale the size of $\mathcal{H}$ down accordingly if this is not the case. We follow similar steps for negative numbers with $\mathcal{L}$. Finally, let $g : \mathbb{R}^d \rightarrow [-1, 1]^d$ be a function that maps returns into position sizing allocations.

We begin by defining S , the magnitude of our asset choices, as

$$S_l = \sum_{\hat{\mathbf{y}}_{t+1}^{j,k} \in \mathcal{H} \cup \mathcal{L}} |\hat{y}_{t+1}^{j,k}|. \quad (14)$$

Next, we define a ‘long and short’ (lns) position sizing strategy as

$$\mathbf{w}_{t+1}^{\text{lns}} = g_{\text{lns}}(\hat{\mathbf{y}}_{t+1}^k, l) = \begin{cases} \frac{\hat{y}_{t+1}^{j,k}}{S_l} & \text{for } \hat{\mathbf{y}}_{t+1}^{j,k} \in \mathcal{H} \cup \mathcal{L} \\ 0 & \text{otherwise.} \end{cases} \quad (15)$$

A long and short strategy has the drawback of requiring us to take both a long and short position, regardless of the magnitude of the forecasts. Instead, we can define a ‘long or short’ (los) strategy that only takes positions on the l ETFs that have forecasts with the *largest magnitude* as

$$\mathbf{w}_{t+1}^{\text{los}} = g_{\text{los}}(\hat{\mathbf{y}}_{t+1}^k, l) = \begin{cases} \frac{\hat{y}_{t+1}^{j,k}}{S_l} & \text{for } \hat{\mathbf{y}}_{t+1}^{j,k} \in \mathcal{B} \\ 0 & \text{otherwise,} \end{cases} \quad (16)$$

where $\mathcal{B}$ is the set of ETFs whose forecasts have the largest magnitude, and S_l is redefined to use $\mathcal{B}$ instead of $\mathcal{H} \cup \mathcal{L}$ in its summation.

Suppose instead, we seek to eliminate drawdowns from short positions. In that case, we may redefine g appropriately to use a ‘long-only instead’ (lo) position sizing strategy by recomputing S_l to use only $\mathcal{H}$ instead of $\mathcal{H} \cup \mathcal{L}$ in its summation. It would have allocations

$$\mathbf{w}_{t+1}^{\text{lo}} = g_{\text{lo}}(\hat{\mathbf{y}}_{t+1}^k, l) = \begin{cases} \frac{\hat{y}_{t+1}^{j,k}}{S_l} & \text{for } \hat{\mathbf{y}}_{t+1}^{j,k} \in \mathcal{H} \\ 0 & \text{otherwise.} \end{cases} \quad (17)$$

As we will see in Section 6, we find that using a long-only strategy is optimal except for periods of economic difficulty when shorts yield substantial returns. To account for this finding, we define one final position sizing strategy that mixes both the long-only strategy and the long or short strategy. It uses a long or short strategy if the following month is forecasted to be in REGIME 0 (the regime associated with economic difficulty), and a long-only strategy for every other regime. This strategy is called the ‘mixed’ (mx) strategy.

5.3.2. Black-Litterman decision criteria. The model proposed by Black and Litterman (1991) combines market equilibrium returns with investors’ own views to generate adjusted

expected returns. In our implementation, instead of using market equilibrium returns derived from betas, we use sample means and covariances as prior beliefs. These are then updated with investor views, which in our case are constructed from regime-conditional expected returns.

For each ETF j , the sample mean μ_j gives the prior expected return, with uncertainty captured by the sample covariance matrix Σ . Using our regime model, the views are derived from the expected returns conditional on the most likely next regime i_{t+1}^* .

For each ETF j , the view on its expected return is

$$q_{j,t+1}^* = \hat{y}_{t+1}^{j,bl} = \mathbb{E} \left[r_{t+1}^j \mid \text{REGIME} = i_{t+1}^* \right] = \hat{\mu}_{j,1:t}^* \quad (18)$$

where $\hat{\mu}_{j,1:t}^*$ is the sample mean return calculated using data from time steps 1 to t where the regime matches i_{t+1}^* . These $q_{j,t+1}^*$ will later be used to build allocations in the Black-Litterman framework. In particular, these values serve as the ‘views’ in the Black-Litterman model.

Given the regime-conditional views $q_{j,t+1}^*$ for each ETF j , we can construct the portfolio weights using the Black-Litterman framework. The posterior distribution of expected returns leads to the following allocation

$$\mathbf{w}_{t+1} = [(\tau \Sigma_{1:t})^{-1} + P_{t+1}^\top \Omega_{t+1}^{-1} P_{t+1}]^{-1} [(\tau \Sigma_{1:t})^{-1} \hat{\boldsymbol{\mu}}_{1:t} + P_{t+1}^\top \Omega_{t+1}^{-1} \mathbf{q}_{t+1}^*], \quad (19)$$

where $\hat{\boldsymbol{\mu}}_{1:t}$ and $\Sigma_{1:t}$ are the sample mean vector and covariance matrix estimated using data from time steps 1 to t , τ is a scaling parameter representing the uncertainty in the previous beliefs, P_{t+1} is the pick matrix expressing which assets have views for time $t+1$, $\mathbf{q}_{t+1}^*$ is the vector of regime-conditional expected returns derived in the previous section, and Ω_{t+1} represents the uncertainty matrix for our regime-conditional views.

5.4. Performance metrics

The performance of different portfolio models is evaluated using several metrics: Sharpe ratio, Sortino ratio, average drawdown (AvgDD), maximum drawdown (MaxDD), and the percentage of positive returns (% Positive Ret.). We present results for portfolios based on two forecasting models: the naive model (Naive) and the linear ridge model (Ridge). For each forecasting model, we analyze four types of decision criteria: long-only (lo), long-and-short (lns), long-or-short (los), and mixed (mx) portfolios, as described in Section 5.3. In addition, we consider two parametric decision criteria that do not directly rely on forecasting model outputs: the mean-variance optimization model (MVO) and the Black-Litterman model (BL). The BL model incorporates regime information through its priors, whereas the MVO does not. Constraints are imposed in the optimization procedures to mimic the long-only and long-and-short decision criteria.

6. Experimental results

6.1. ETFs experiment

We begin our experimental setup with the FRED-MD dataset, which spans from February 2000 to January 2023, as detailed in Section 4. This macroeconomic dataset is merged with beginning-of-the-month returns for ten Exchange-Traded Funds (ETFs), described in table 2, sourced from the Wharton Research Data Services (WRDS). The ETF dataset covers the period from February 2000 to December 2022.

After merging the two datasets, the final dataset comprises 746 monthly observations spanning the period from 2000 to 2022. For our analysis, we define an estimation window of 48 months (four years) for both the regime detection and prediction models. Using this estimation window, we implement a fixed window, walk-forward prediction exercise, where predictions are made one month ahead. We use these predictions to build portfolios as defined in Section 5.3.

Figures 11, 12, and 13 present comparative boxplots of performance metrics for control (random regimes) versus treatment (non-random regimes) across our three modeling approaches. Each figure displays four key metrics: Sharpe ratio, Sortino ratio, maximum drawdown (MaxDD), and percentage of positive returns. The boxplots are accompanied by p-values from t-tests for two related samples, where the null hypothesis states that the control mean is equal to or greater than the treatment mean. Table 3 complements these visualizations by providing statistical evidence through the Nemenyi test, quantifying the significance of differences between random and non-random regime approaches using rank statistics.

In table 3, *Control Rank* and *Treatment Rank* report the average ordinal scores assigned by the Nemenyi post-hoc test to the baseline (random-regime) and proposed (non-random-regime) methods, respectively, for each performance metric. Ranks are defined so that a larger average rank corresponds

Table 2. Economic characterizations of ETFs used in this work.

ETF Ticker	Sector	Description
SPY	S&P 500	Tracks the largest 500 U.S. companies.
XLB	Materials	Covers chemicals, mining, and construction materials.
XLE	Energy	Includes oil, gas, and renewable energy companies.
XLF	Financials	Banks, insurers, and asset managers.
XLI	Industrials	Aerospace, defense, and manufacturing.
XLK	Technology	Software, hardware, and IT services.
XLP	Staples	Essential goods like food and household products.
XLU	Utilities	Electricity, water, and gas providers.
XLV	Healthcare	Pharmaceuticals, biotech, and providers.
XLY	Discretionary	Retail, automotive, and entertainment.

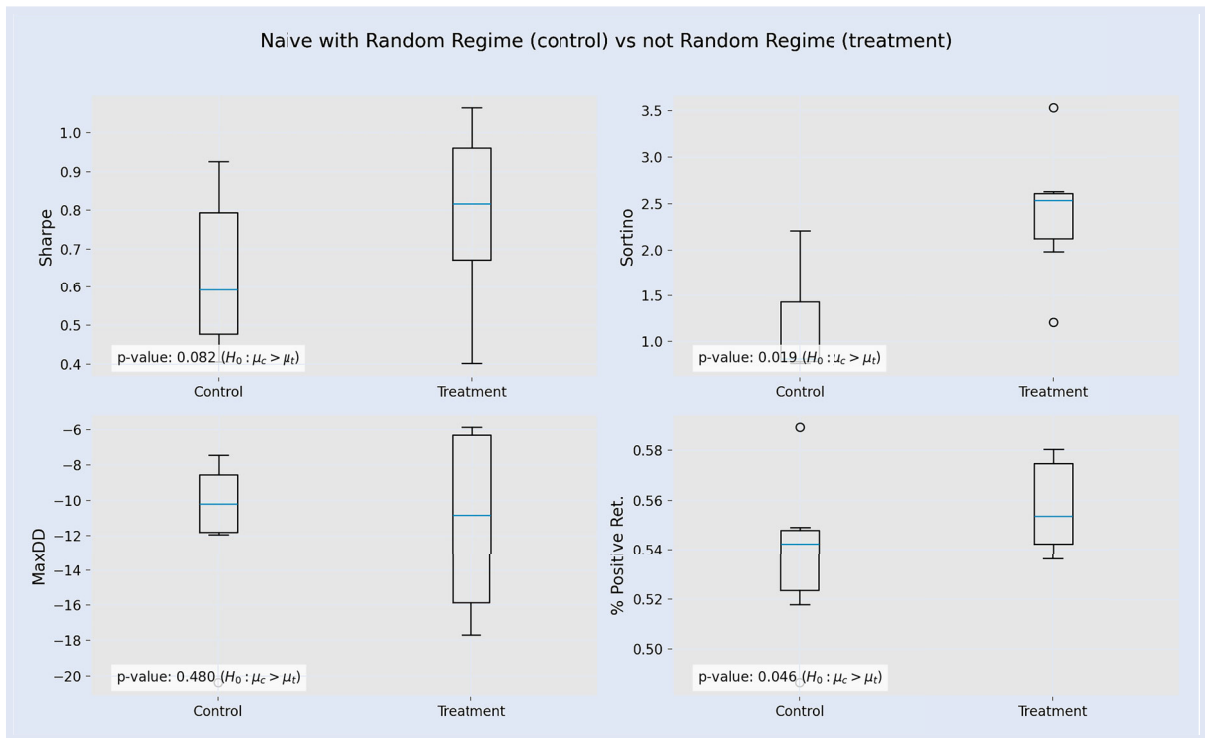


Figure 11. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the naive method based on random regimes, whereas the treatment is defined as the naive method based on non-random regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

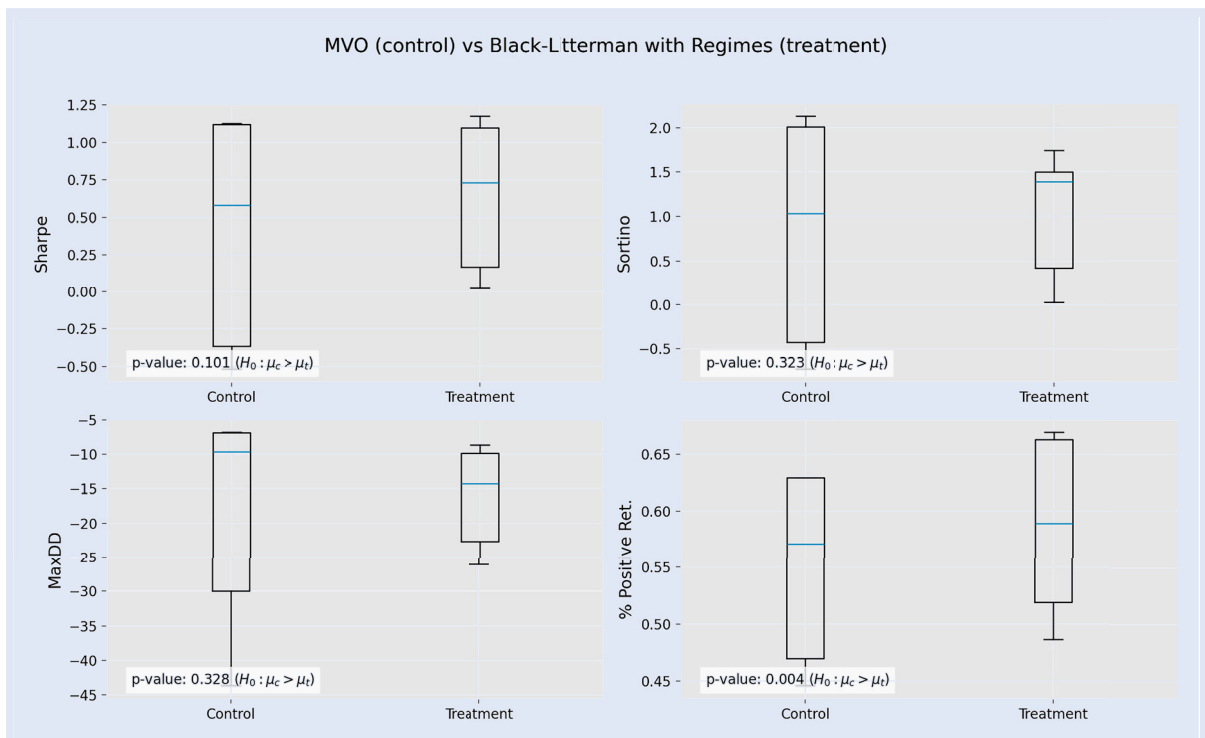


Figure 12. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the mean-variance optimization method (MVO), whereas the treatment is defined as the Black-Litterman (BL) approach where the views are defined based on the regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

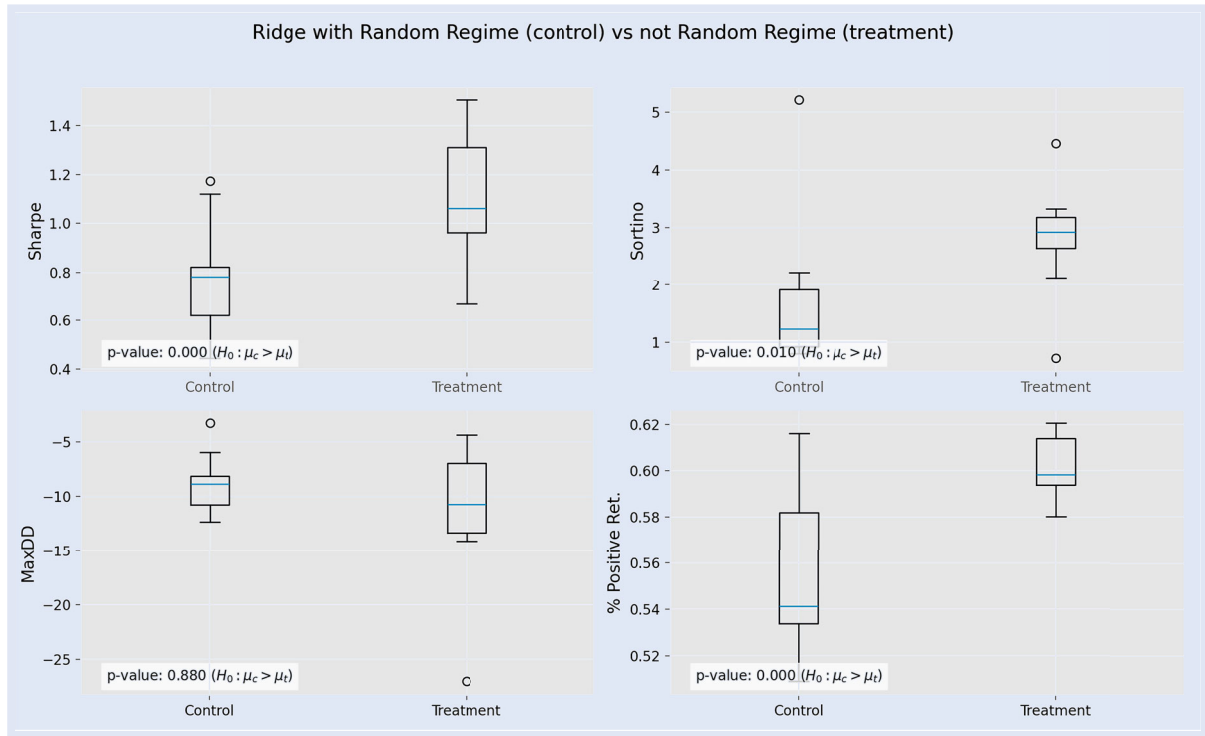


Figure 13. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the ridge (Ridge) method based on random regimes, whereas the treatment is defined as the ridge method based on non-random regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

Table 3. The table shows results for the Nemenyi test.

Metric	Control Rank	Treatment Rank	p-value
Panel A: Naive Random vs not Random Regimes Portfolios			
Sharpe	1.333	1.667	0.082*
Sortino	1.167	1.833	0.019**
MaxDD	1.500	1.500	0.480
% Positive Ret.	1.250	1.750	0.046**
Panel B: BL Random vs not Random Regimes Portfolios			
Sharpe	1.333	1.667	0.068*
Sortino	1.500	1.500	0.295
MaxDD	1.667	1.333	0.394
% Positive Ret.	1.083	1.917	0.003***
Panel C: Ridge Random vs not Random Regimes Portfolios			
Sharpe	1.000	2.000	0.000***
Sortino	1.167	1.833	0.010***
MaxDD	1.583	1.417	0.880
% Positive Ret.	1.083	1.917	0.000***

Note: Control and treatment methods are defined as in figures 11–13. The table displays the rank statistics for control and treatment methods, along with the associated p-values.

***, **, * indicate significance at 1%, 5%, and 10% levels

to better performance. The p-values indicate the probability of observing such a rank difference under the null hypothesis that there is no true difference between the two methods.

The naive forecasting method with non-random regimes (figure 11) generally performs better and is more stable on the appropriate metrics, particularly for Sharpe and Sortino ratios. This visual pattern is confirmed by Panel A in table 3, which shows statistically significant improvements in the Sortino

ratio (p-value = 0.019) and percentage of positive returns (p-value = 0.046), with a marginal improvement in the Sharpe ratio (p-value = 0.082).

Figure 12 compares MVO (control) and Black-Litterman with regime-based views (treatment). The boxplots reveal less pronounced differences than the naive approach, except for the percentage of positive returns. This observation aligns with Panel B of table 3, where the most significant improvement appears in the percentage of positive returns (p-value = 0.003). At the same time, other metrics show more modest or insignificant differences.

The linear ridge regression results (figure 13) demonstrate the most striking improvements when using non-random regimes. The treatment group exhibits significantly higher and more concentrated Sharpe ratios and Sortino ratios, as well as a higher percentage of positive returns, as shown in Panel C of table 3. Statistical results present substantial evidence to reject both null hypotheses for both metrics. Regarding the t-test, we estimate p-values of 0.004, 0.01, and 0.013 for Sharpe, Sortino, and percentage of positive returns, respectively. This means that we can strongly reject the null hypothesis that the average metric for the ridge model using random regimes is higher than that for the one not using random regimes. Similar evidence is found using the Nemenyi test, where we rejected the null hypothesis that there is no difference between the two models in terms of Sharpe, Sortino, and percentage of positive returns.

These findings suggest that structured regime information consistently improves the quality of returns across all three approaches, with the most substantial benefits observed in the linear ridge regression framework. The improvement in Sharpe ratios and the percentage of positive returns indicates

Table 4. Performance metrics for portfolio strategies with naive and MVO approaches.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
naive_lns_2	0.651	1.984	− 14.942	0.580
naive_mx_2	0.773	1.771	− 14.643	0.585
naive_los_2	0.796	1.742	− 14.595	0.580
naive_lo_2	0.981	2.532	− 6.163	0.540
naive_lns_3	0.402	3.533	− 16.148	0.558
naive_mx_3	0.740	1.948	− 14.681	0.576
naive_los_3	0.568	2.357	− 14.773	0.558
naive_lo_3	0.899	2.630	− 5.859	0.536
naive_mx_4	0.746	1.486	− 17.807	0.589
naive_lns_4	0.734	1.207	− 17.699	0.580
naive_los_4	0.788	1.618	− 17.064	0.580
naive_lo_4	1.065	2.541	− 6.719	0.549
mvo_lns_2	− 0.516	− 0.601	− 35.806	0.455
mvo_lo_2	1.128	2.022	− 6.776	0.629
mvo_lns_3	0.058	0.102	− 12.390	0.513
mvo_lo_3	1.104	1.963	− 6.846	0.629
mvo_lns_4	− 0.510	− 0.723	− 43.622	0.446
mvo_lo_4	1.129	2.132	− 6.933	0.629
spy	0.818	1.331	− 33.492	0.662
ew	0.838	1.445	− 32.286	0.662

Note: Bold values indicate the best performance per column.

that regime information helps models better capture market opportunities.

Second, the pattern of improvements suggests that regime information primarily enhances return generation capabilities rather than downside risk management, as evidenced by the generally insignificant changes in maximum drawdown across all approaches.

Third, the varying degrees of improvement across methods indicate that more sophisticated models, particularly the linear ridge regression, may be better equipped to leverage regime information effectively. This suggests that the choice of the modeling framework plays a crucial role in maximizing the benefits of regime-based information in portfolio management.

To complement the results of the previous hypothesis testing exercise, Tables 4–6 and Figures 14–17 evaluate portfolio performance using standard portfolio metrics and cumulative returns. The tables present detailed summary statistics for all methods tested, while the figures focus on the single best-performing variant (in terms of Sharpe ratio) from each group to facilitate visual comparison.

Examining Figures 14–17 and Tables 4–6, we observe several notable performance patterns. Among naive strategies, naive_lo_4 achieves a Sharpe ratio of 1.065 and exhibits the lowest maximum drawdown (− 6.719) within its category. The best naive portfolios consistently outperform the SPY benchmark and the MVO during major market downturns, as shown in figure 15. We see in that same figure that although the naive long-only portfolio (blue) successfully mitigates drawdowns in 2008, it does not keep pace with SPY during the rapid post-COVID rebound. From March 2020 through December 2022, SPY's cumulative return remains above all naive variants, indicating that regime-conditioned Sharpe-based forecasts alone could not have accurately timed

Table 5. Performance metrics for portfolio strategies with Black-Litterman and MVO approaches. Bold values indicate the best performance per column.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
bl_lns_2	0.390	1.507	− 17.926	0.522
bl_lo_2	1.177	1.744	− 8.636	0.665
bl_lns_3	0.023	0.023	− 26.081	0.518
bl_lo_3	1.071	1.306	− 10.612	0.670
bl_lns_4	0.096	0.130	− 24.185	0.487
bl_lo_4	1.107	1.479	− 9.620	0.656
mvo_lns_2	− 0.516	− 0.601	− 35.806	0.455
mvo_lo_2	1.128	2.022	− 6.776	0.629
mvo_lns_3	0.058	0.102	− 12.390	0.513
mvo_lo_3	1.104	1.963	− 6.846	0.629
mvo_lns_4	− 0.510	− 0.723	− 43.622	0.446
mvo_lo_4	1.129	2.132	− 6.933	0.629
spy	0.818	1.331	− 33.492	0.662
ew	0.838	1.325	− 32.286	0.662

Table 6. Performance metrics for linear ridge regression and MVO approaches.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
ridge_lns_2	0.666	0.718	− 27.020	0.621
ridge_mx_2	1.015	2.865	− 12.300	0.612
ridge_los_2	0.876	2.127	− 9.926	0.621
ridge_lo_2	1.134	4.449	− 4.389	0.607
ridge_lns_3	0.953	2.977	− 13.266	0.594
ridge_mx_3	1.330	2.674	− 7.795	0.598
ridge_los_3	1.051	3.322	− 10.976	0.621
ridge_lo_3	1.505	3.170	− 4.386	0.594
ridge_lns_4	0.966	3.067	− 14.090	0.580
ridge_mx_4	1.303	2.554	− 10.470	0.589
ridge_los_4	1.074	2.813	− 13.546	0.598
ridge_lo_4	1.490	3.201	− 4.368	0.594
mvo_lns_2	− 0.516	− 0.601	− 35.806	0.455
mvo_lo_2	1.128	2.022	− 6.776	0.629
mvo_lns_3	0.058	0.102	− 12.390	0.513
mvo_lo_3	1.104	1.963	− 6.846	0.629
mvo_lns_4	− 0.510	− 0.723	− 43.622	0.446
mvo_lo_4	1.129	2.132	− 6.933	0.629
spy	0.818	1.331	− 33.492	0.662
ew	0.838	1.325	− 32.286	0.662

Note: Bold values indicate the best performance per column.

the swift market recovery. Looking at table 4, among all naive sizing schemes, the long-only $l = 4$ variant achieves the highest Sharpe (1.065) and lowest MaxDD (− 6.719). However, as figure 15 shows, its absolute cumulative return still trails that of SPY in the entire sample, demonstrating that improved risk-adjusted metrics did not translate into outright outperformance of the benchmark.

In the Black-Litterman implementations, bl_lo_2 records the highest Sharpe ratio (1.177) and the superior positive return percentage (0.665) compared to MVO strategies (0.629), yet its cumulative path (figure 16) is still close to the MVO benchmark. However, MVO strategies achieve better Sortino ratios, with mvo_lo_4 reaching 2.132. This suggests that regime-informed views effectively dampen downside but yield only modest alpha relative to standard MVO

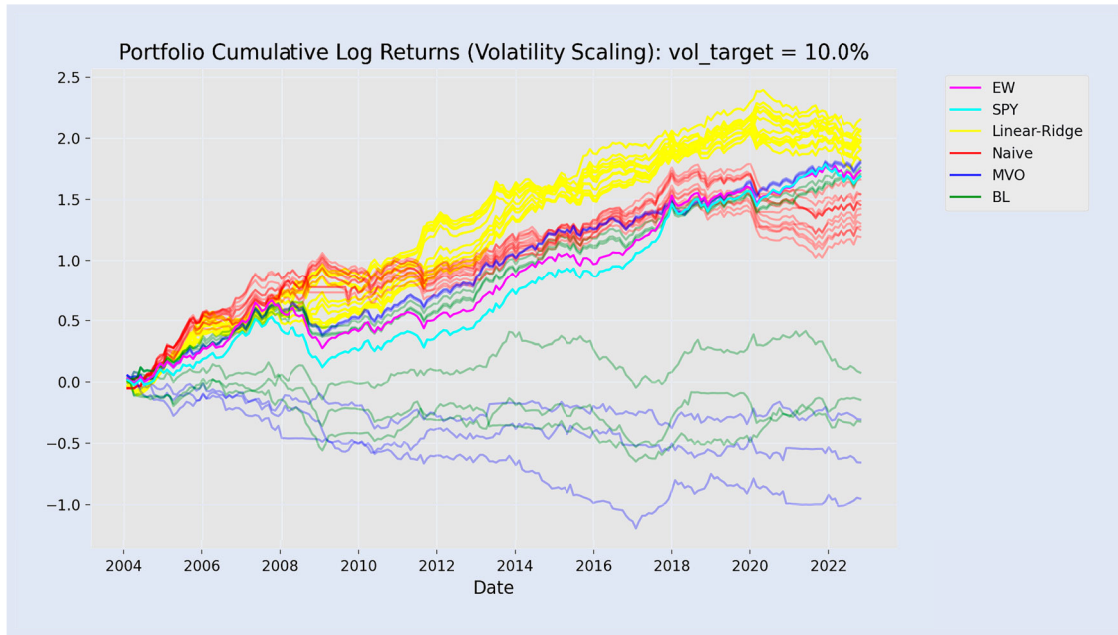


Figure 14. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Linear ridge strategies are in yellow, naive strategies are in red, mean-variance optimization strategies are in blue, and Black-Litterman strategies are in green.

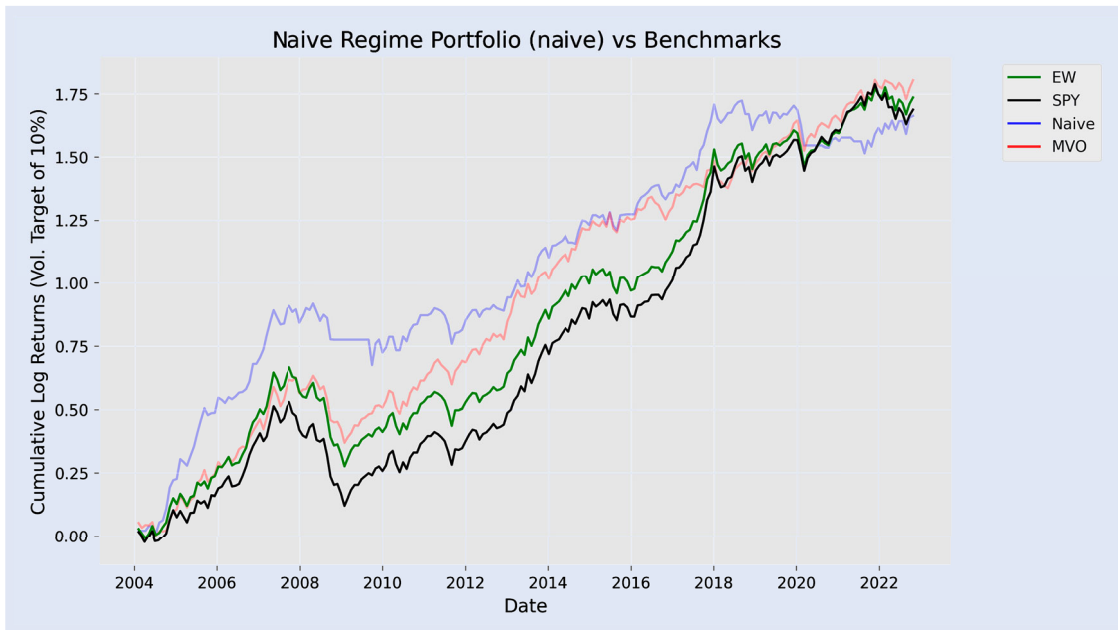


Figure 15. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Naive strategies are represented in blue, while mean-variance optimization strategies are represented in red.

benchmarks. Figure 16 shows BL strategies tracking closer to market benchmarks than MVO approaches, while long-short variants of both methods frequently produce negative Sharpe ratios (e.g. `mvo_lns_2` at -0.516). The figure also shows that Black-Litterman and MVO portfolios (in pink) outperformed SPY during the 2008 market drawdown. However, by the end of the sample, their cumulative performance converges to levels similar to SPY. Even with regime-based views incorporated, these allocations remain conservative. SPY outperforms the Black-Litterman portfolio and performs similarly to the MVO strategy during the 2020–2023 period, underscoring that regime-driven priors

alone do not guarantee excess returns over a passive equity benchmark.

Again, the linear ridge regression approach produces the strongest overall metrics. The `ridge_lo_3` configuration achieves the highest Sharpe ratio (1.505) among all strategies, while `ridge_lo_2` demonstrates exceptional downside protection with a Sortino ratio of 4.449 and the lowest maximum drawdown (-4.389) among all approaches. Figure 17 shows that the ridge regression strategies (blue) decisively outperform SPY, going ahead in the early 2010s and widening their lead through the post-COVID rally. The strongest divergence occurs between 2020 and 2021, illustrating that

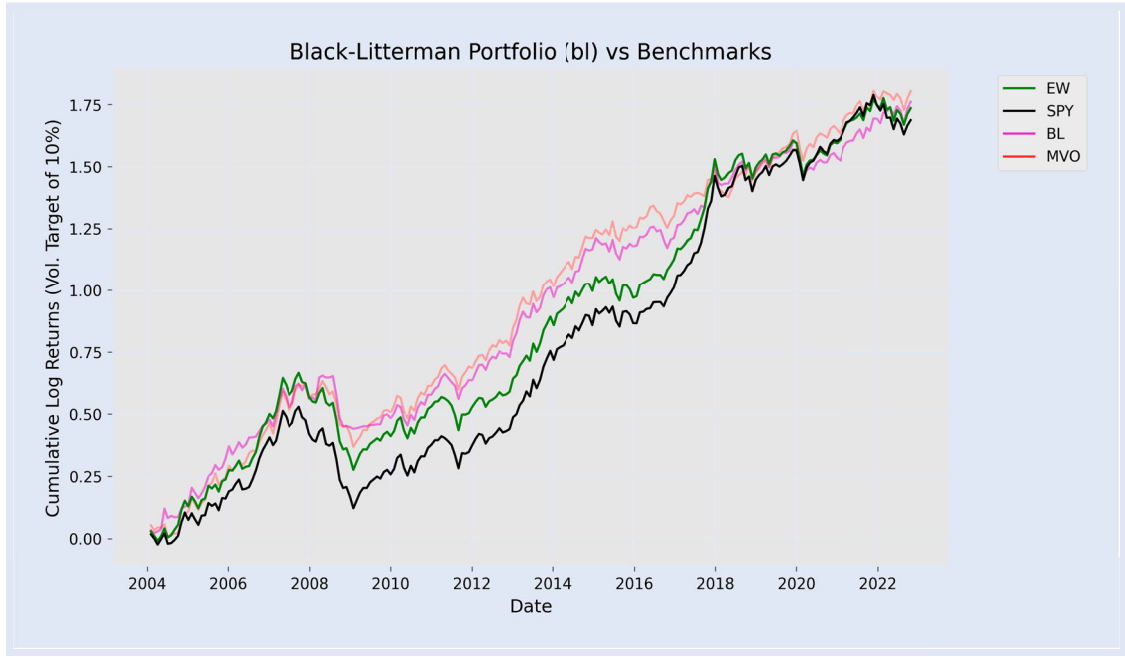


Figure 16. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Black-Litterman strategies are in pink, and mean-variance optimization strategies are in red.

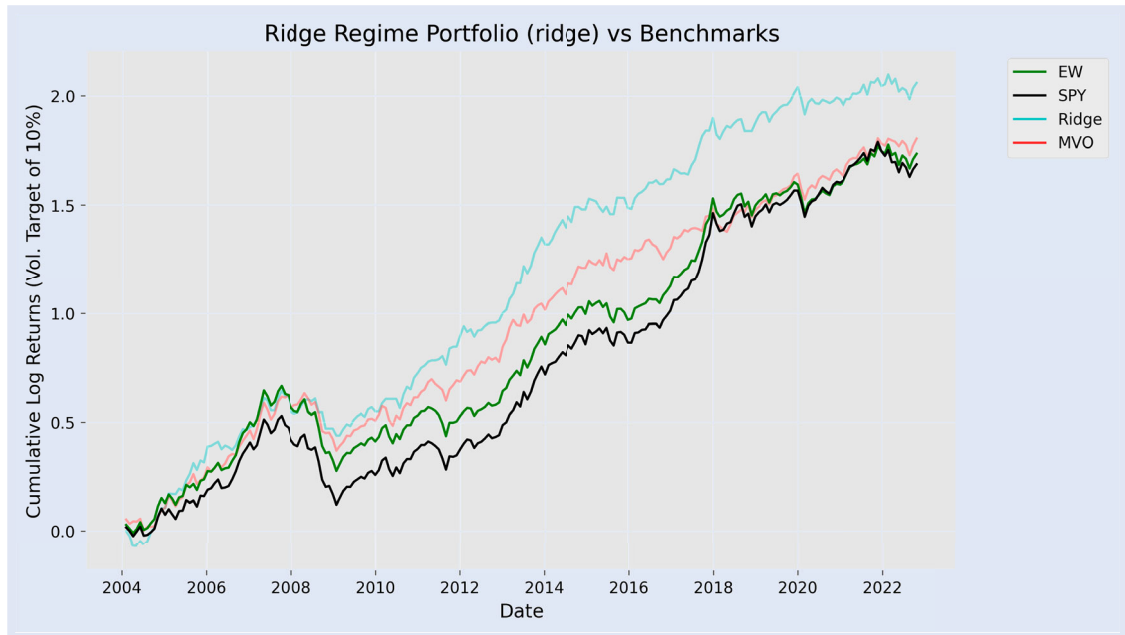


Figure 17. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Linear ridge strategies are in blue, and mean-variance optimization strategies are in red.

regularized learning coupled with regime signals can capture regime shifts and exploit market recoveries more effectively than benchmark allocations. Because ridge long-only portfolios (particularly $l=2,3$) achieve Sharpes above 1.4 and MaxDDs below 5%, which far exceeds both SPY and EW, we can conclude that their superior risk adjusted metrics, combined with the cumulative gains shown in figure 17, confirm that the ridge model most fully translates regime information into consistent outperformance.

Our results underscore both the challenge and the potential of active portfolio management. The benchmark indices (SPY and EW) demonstrate remarkably consistent performance

with 66.2% positive returns, highlighting the difficulty in outperforming passive strategies. This robust benchmark performance makes the success of certain regime-based strategies particularly notable.

The linear ridge regression approach, especially in its long-only implementations, stands out by achieving superior risk-adjusted returns. With a Sharpe ratio of 1.505 and exceptional downside protection, these results suggest that machine learning approaches may be particularly well suited to capture regime dynamics. Their performance is especially meaningful given the high hurdle set by market benchmarks.

A striking pattern across all strategies is the superior performance of long-only implementations compared to their long-short counterparts. This finding challenges conventional wisdom on the diversification benefits of short positions, suggesting that the added complexity may introduce more noise than signal in regime-based strategies. Similarly, lower-dimensional models ($l=2$ or $l=3$) consistently outperform higher-dimensional ones ($l=4$), indicating that simpler specifications may better capture fundamental market dynamics.

Although the Black-Litterman approach does not achieve the highest absolute performance, it demonstrates an interesting balance between active management and benchmark tracking. Its tendency to align closely with market benchmarks while still achieving competitive Sharpe ratios suggests potential value for investors seeking enhanced passive strategies.

These findings carry significant implications for practical implementation. While outperforming market benchmarks remains a formidable challenge, our results suggest that

thoughtfully implemented regime-based strategies, particularly those utilizing machine learning approaches, can achieve this goal. The key element is not to increase model complexity but to make informed choices about implementation, favoring simpler, long-only approaches that focus on capturing fundamental market dynamics. While not universal across all configurations, this success in outperforming challenging benchmarks demonstrates the potential value of regime-based approaches in portfolio management.

6.2. Futures experiment

We continue our experimental setup again with the FRED-MD dataset, but now this macroeconomic dataset is merged with beginning-of-the-month returns for eight selected futures time series, described in table 7, sourced from Pinnacle Data. The futures dataset covers the period from January 2000 to December 2023.

Rather than including all futures directly in the modeling—an approach that would introduce redundancy and increase noise—we employ a systematic procedure to select a representative subset of instruments across asset classes. This is particularly important for the forecasting models, which do not explicitly model inter-asset correlations. Moreover, for models such as the mean variance optimizer, well-documented challenges related to the estimation of covariance matrices can result in unstable or overfit portfolios (López de Prado 2016); addressing such issues is beyond the scope of this article.

To construct a more diversified and parsimonious input set, we first compute a correlation matrix based on the standardized returns of the entire future panel. We then apply KMeans clustering with $k=4$, a number selected to broadly capture distinct asset class groupings (e.g. equities, rates, commodities, and currencies). The resulting cluster assignments are reported in table 8.

Within each cluster, we apply Principal Component Analysis (PCA) to the matrix of futures returns, extracting the

Table 7. Descriptions of asset tickers used in this work, grouped by asset class.

Ticker	Category	Description
ZC	Grains	Corn
ZO	Grains	Oats
ZL	Grains	Soybean Oil
ZR	Grains	Rough Rice
ZW	Grains	Wheat
CC	Softs	Cocoa
DA	Softs	Milk
JO	Softs	Orange Juice
KC	Softs	Coffee
SB	Softs	Sugar
LB	Livestock	Lumber
ZF	Livestock	Feeder Cattle
ZZ	Livestock	Lean Hogs
ZT	Livestock	Live Cattle
ZG	Metals	Gold
ZI	Metals	Silver
ZK	Metals	Copper
ZP	Metals	Platinum
ZN	Energy	Natural Gas
ZU	Energy	Crude Oil
CB	Bonds	10-Year Canadian Bond
DT	Bonds	Euro Bund
EC	Bonds	Eurodollar
FB	Bonds	5-Year U.S. Treasury Bond
GS	Bonds	U.K. Gilt
TU	Bonds	2-Year U.S. Treasury Note
TY	Bonds	10-Year U.S. Treasury Note
UB	Bonds	Ultra U.S. Treasury Bond
UZ	Bonds	Euro Schatz
AN	FX	AUD/USD
CN	FX	CAD/USD
BN	FX	GBP/USD
DX	FX	U.S. Dollar Index
JN	FX	JPY/USD
MP	FX	MXN/USD
SN	FX	CHF/USD
NK	Equities	Nikkei 225
ES	Equities	E-mini S&P 500
LX	Equities	FTSE 100
XU	Equities	Euro Stoxx 50

Table 8. Futures tickers grouped by KMeans cluster ($k=4$).

Cluster 0	Cluster 1	Cluster 2	Cluster 3
XU (Equities)	DT (Bonds)	AN (FX)	DX (FX)
ES (Equities)	CB (Bonds)	CN (FX)	ZC (Grains)
NK (Equities)	GS (Bonds)	ZG (Metals)	ZT (Livestock)
MP (FX)	TU (Bonds)	ZI (Metals)	ZZ (Livestock)
LX (Equities)	TY (Bonds)	ZK (Metals)	ZF (Livestock)
	EC (Bonds)	ZP (Metals)	LB (Livestock)
	UB (Bonds)	BN (FX)	SB (Softs)
	JN (FX)	SN (FX)	KC (Softs)
	UZ (Bonds)		JO (Softs)
	FB (Bonds)		DA (Softs)
			CC (Softs)
			ZW (Grains)
			ZR (Grains)
			ZL (Grains)
			ZO (Grains)
			ZN (Energy)
			ZU (Energy)

Note: Categories are shown in parentheses.

Table 9. Selected futures tickers per cluster, based on PCA loading correlations with the first principal component.

Cluster	Selected Tickers
Cluster 0	XU, ES
Cluster 1	TY, FB
Cluster 2	ZI, ZG
Cluster 3	ZN, ZU

first three principal components. We then compute the Pearson correlation coefficient between each asset's return time series and the loadings of the first principal component (PC1). Assets are ranked in descending order by the absolute value of their correlation with PC1, and we select the top two assets from each cluster. This procedure ensures that we retain the most representative and informative assets from each cluster, without relying on the entire panel.

The final list of selected futures contracts used for modeling is shown in table 9.

Figures 18, 19, and 20 present comparative boxplots of portfolio performance metrics for control (random regime-based signals) versus treatment (non-random regime-based signals) across the naive, mean-variance optimization (MVO), and ridge-based forecasting strategies, respectively. Across all figures, we display the Sharpe ratio, Sortino ratio, maximum drawdown (MaxDD), and the proportion of positive returns, along with p-values from paired t-tests testing the null

hypothesis that the control mean is greater than the treatment mean.

The treatment portfolios based on non-random regime signals show meaningful improvements in the naive and ridge models. For the naive model (figure 18), treatment portfolios yield visibly higher median Sharpe and Sortino ratios, with lower dispersion, along with improved performance in maximum drawdown and the frequency of positive returns. These differences are statistically significant, as reflected by low p-values (e.g. Sharpe $p = 0.002$, % positive returns $p = 0.012$). In contrast, the MVO vs. Black-Litterman comparison (figure 19) reveals negligible visual and statistical differences in all metrics, suggesting that the regime views incorporated into BL have a minimal impact relative to traditional MVO in this setting. The most striking gains are evident in the comparison of the ridge model (figure 20), where non-random regimes yield substantially higher and tighter distributions across all return-based metrics, including a notable improvement in positive return frequency and a reduction in drawdowns. These results confirm that regime information, when structured and non-random, enhances performance, particularly in models like ridge regression that can effectively absorb such signals.

Table 10 presents the Nemenyi test results, which reinforce the boxplot findings across the naive, Black-Litterman, and ridge models. Panel A shows that the naive model benefits significantly from non-random regime conditioning, with Sharpe, Sortino, MaxDD, and % positive returns all improving—consistent with the patterns in figure 18. These results confirm that even simple models can gain structure and performance from regime signals.

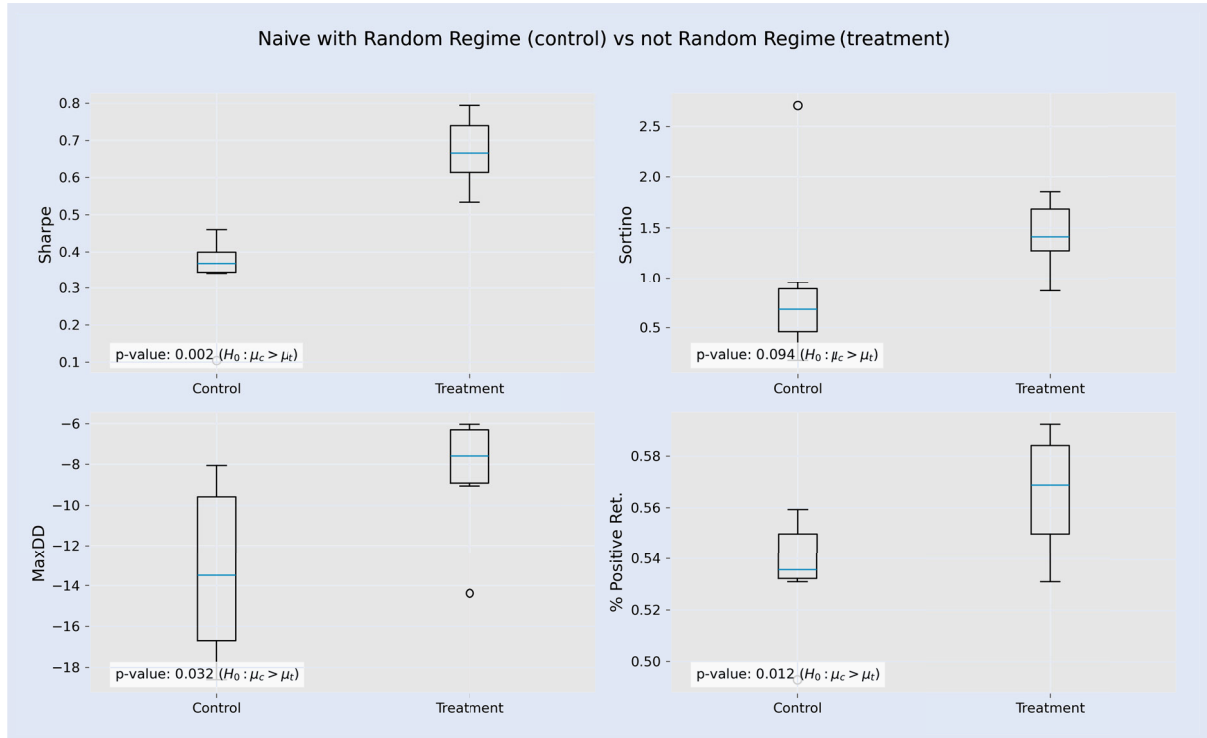


Figure 18. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the naive method based on random regimes, whereas the treatment is defined as the naive method based on non-random regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

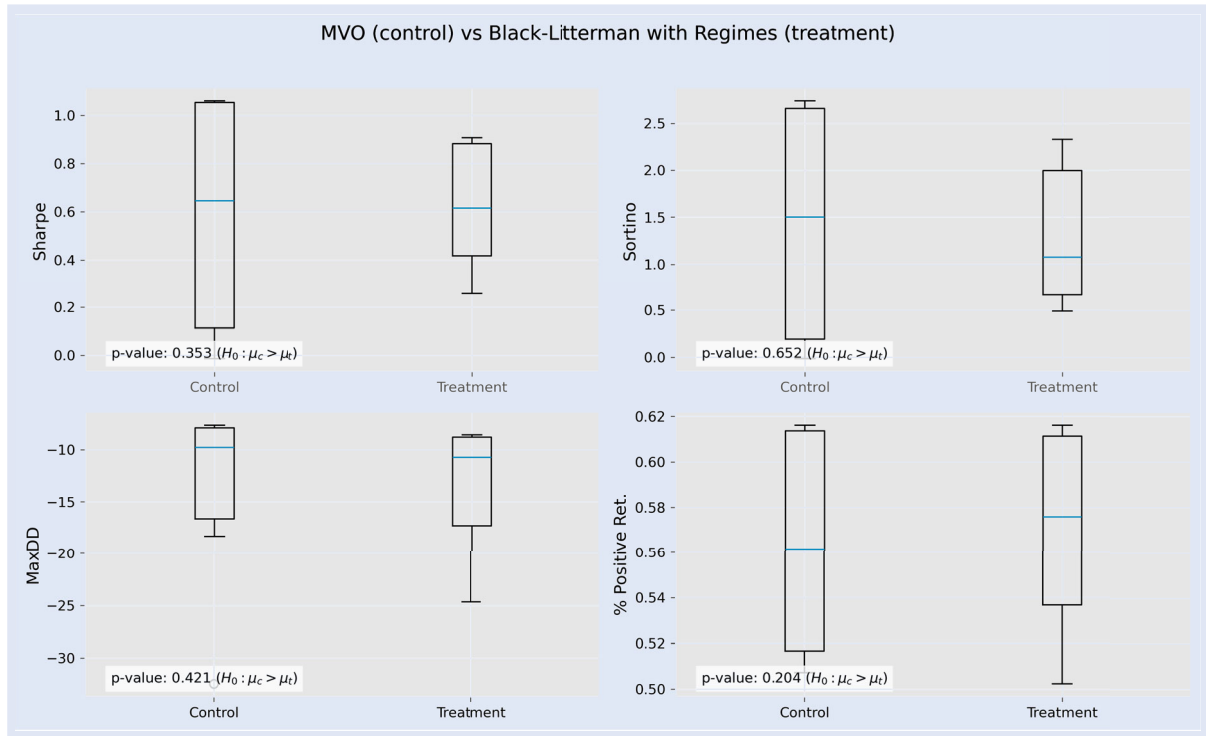


Figure 19. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the mean-variance optimization method (MVO), whereas the treatment is defined as the Black-Litterman (BL) approach where the views are defined based on the regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

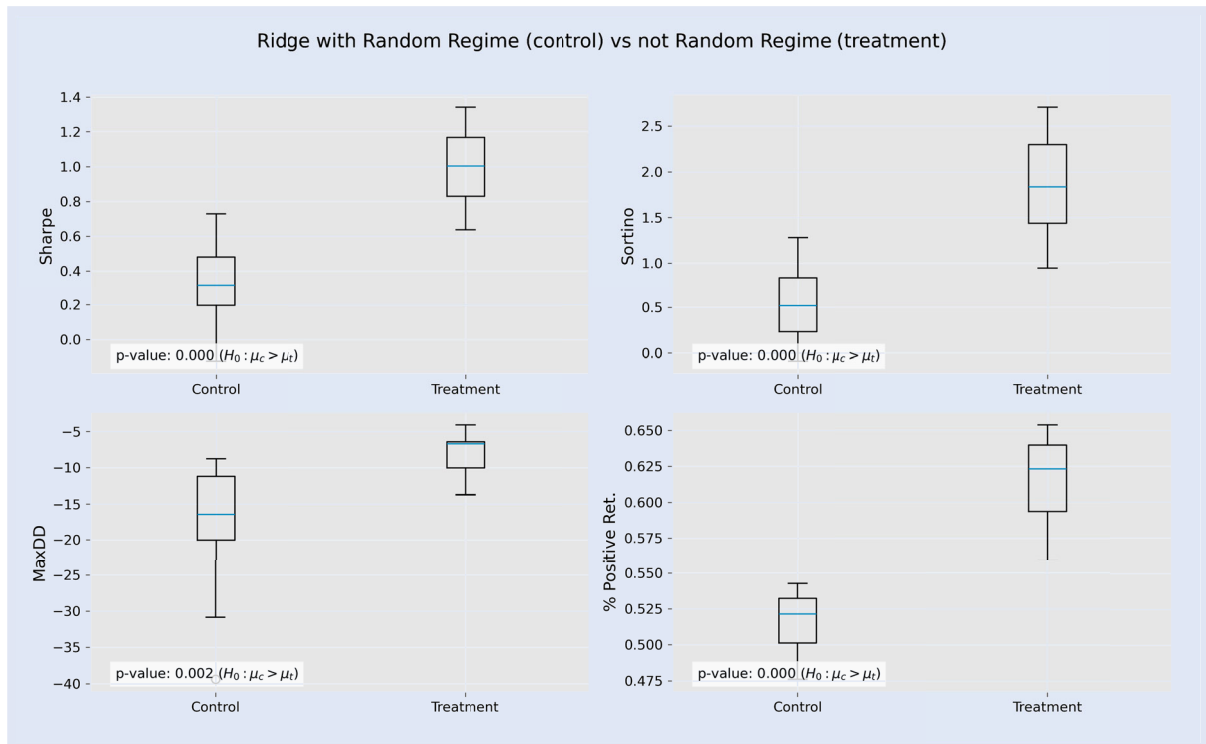


Figure 20. The figure depicts boxplots for control and treatment portfolio metrics. The control is defined as the ridge (Ridge) method based on random regimes, whereas the treatment is defined as the ridge method based on non-random regimes. The figure also shows p-values for a paired t-test (two related samples), where the null hypothesis is that the control mean is higher than the treatment mean.

Table 10. The table shows results for the Nemenyi test. Control and treatment methods are defined as in figure 18–20.

Metric	Control rank	Treatment rank	p-value
Panel A: Naive Random vs not Random Regimes Portfolios			
Sharpe	1.000	2.000	0.002***
Sortino	1.167	1.833	0.094*
MaxDD	1.333	1.667	0.032**
% Positive Ret.	1.167	1.833	0.012**
Panel B: BL Random vs not Random Regimes Portfolios			
Sharpe	1.667	1.333	0.578
Sortino	1.667	1.333	0.628
MaxDD	1.333	1.667	0.703
% Positive Ret.	1.833	1.167	0.984
Panel C: Ridge Random vs not Random Regimes Portfolios			
Sharpe	1.000	2.000	0.000***
Sortino	1.000	2.000	0.000***
MaxDD	1.083	1.917	0.002***
% Positive Ret.	1.000	2.000	0.000***

Note: The table displays the rank statistics for the control and treatment methods, along with the corresponding p-values. ***, **, * indicate significance at 1%, 5%, and 10% levels

Panel B shows no significant improvements for the Black-Litterman approach, aligning with the muted effects observed in figure 19. This suggests that regime information may offer limited marginal value when prior views are already embedded in the model.

Panel C provides the strongest support for regime-based modeling, as the ridge approach shows significant gains across all metrics ($p < 0.01$), echoing the clear advantages observed in figure 20. These results underscore the ability of the ridge model to capitalize on structured regime information, highlighting its effectiveness for regime-aware portfolio construction in futures markets.

Table 11. Performance metrics for portfolio strategies with naive and MVO approaches.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
naive_ins_2	0.596	1.852	− 8.496	0.564
naive_mx_2	0.524	0.871	− 9.476	0.555
naive_los_2	0.528	1.016	− 8.501	0.545
naive_lo_2	0.663	1.358	− 6.697	0.545
naive_ins_3	0.668	1.241	− 9.075	0.573
naive_mx_3	0.954	2.067	− 4.939	0.611
naive_los_3	0.780	1.734	− 4.418	0.573
naive_lo_3	0.794	1.460	− 6.025	0.592
naive_mx_4	0.879	2.462	− 5.101	0.583
naive_ins_4	0.533	0.866	− 14.378	0.531
naive_los_4	0.634	1.526	− 8.087	0.536
naive_lo_4	0.764	1.755	− 6.181	0.588
mvo_ins_2	0.231	0.304	− 19.428	0.536
mvo_lo_2	1.042	2.132	− 7.844	0.616
mvo_ins_3	0.118	0.199	− 17.689	0.521
mvo_lo_3	1.091	3.558	− 7.721	0.607
mvo_ins_4	0.418	0.444	− 17.344	0.555
mvo_lo_4	1.092	3.383	− 7.633	0.611
ew	0.406	0.587	− 34.236	0.582

Note: Bold values indicate the best performance per column.

The final experiment compares the performance of regime-augmented methods against benchmark approaches, namely mean-variance optimization (MVO) and equal-weighted (EW) portfolios. As in the ETF setting, multiple model variations are tested to ensure robustness and avoid cherry-picking. Performance is assessed using four key metrics: the Sharpe ratio, the Sortino ratio, maximum drawdown (MaxDD), and the percentage of positive return days. Figure 21 provides an overview of the cumulative log returns of the best-performing variant from each strategy category on the futures dataset, while Tables 11–13 and Figures 22–24 report detailed

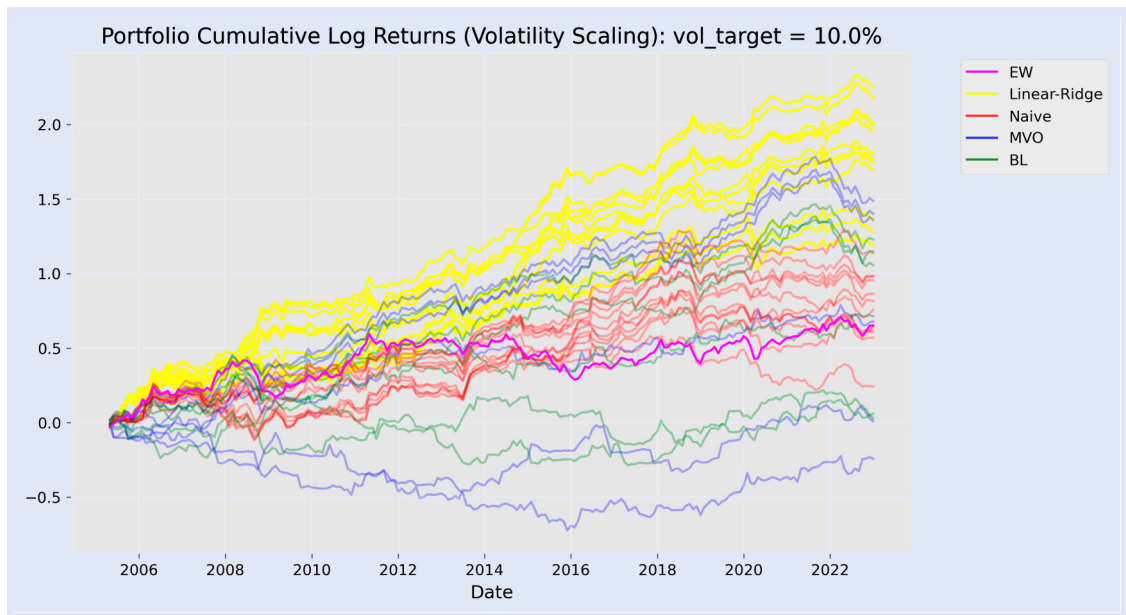


Figure 21. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Linear ridge strategies are in yellow, naive strategies are in red, mean-variance optimization strategies are in blue, and Black-Litterman strategies are in green.

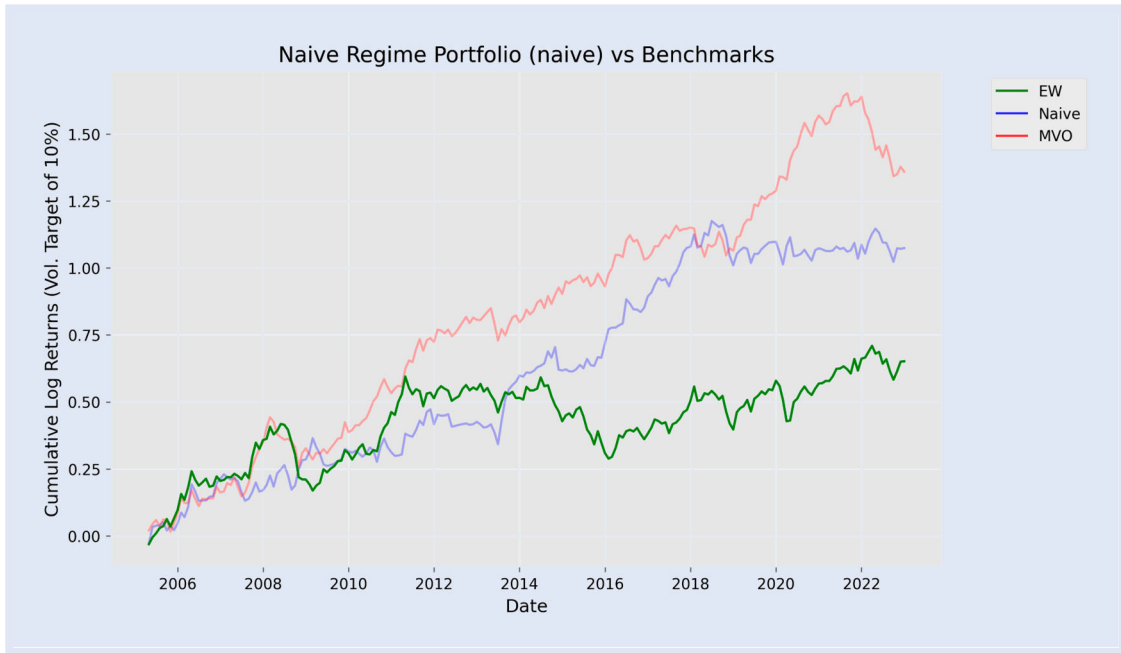


Figure 22. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Naive strategies are represented in blue, while mean-variance optimization strategies are represented in red.

performance metrics and pairwise comparisons. Table 11 and Figure 22 show mixed results when comparing naive models to MVO. While MVO outperforms on risk-adjusted metrics, with the highest Sharpe (1.092 for mvo_lo_4) and Sortino (3.558 for mvo_lo_3) ratios, the naive portfolios exhibit lower drawdowns, particularly naive_mx_3 with a MaxDD of -4.939 . In terms of positive return frequency, both methods yield strong results: MVO reaches a peak of 0.616 (mvo_lo_2), slightly above the best naive portfolio (naive_mx_3 and naive_lo_4, both at 0.611). These findings highlight a trade-off between return stability and downside protection in all modeling choices.

Table 12 presents performance metrics for Black-Litterman (BL) strategies, all of which incorporate regime information through their priors, compared to benchmarks that do not use regimes—namely, MVO and equal-weighted (EW) portfolios. The results indicate that certain BL configurations deliver competitive or even superior performance. For example, the bl_lo_4 model achieves the highest Sharpe ratio (1.210), the lowest maximum drawdown (-6.286), and a strong Sortino ratio (3.492), suggesting that well-constructed regime priors can meaningfully enhance the BL framework. In contrast, long-and-short BL models consistently underperform, often trailing both the MVO and EW strategies, underscoring the sensitivity of BL to the quality and stability of its priors.

While the best MVO configuration (mvo_lo_4) slightly surpasses bl_lo_4 in terms of the percentage of positive return days (0.611 vs. 0.607), more extensive evidence supports the potential of regime-aware BL models, particularly when paired with portfolio constraints such as long-only positioning. Figure 23 shows the cumulative returns for the best models in each class, further confirming that both BL and MVO can outperform the EW benchmark. However, the BL model does not significantly outperform MVO. This mirrors the pattern observed in the ETF experiment, suggesting that structural similarities, such as sensitivity to matrix inversion

Table 12. Performance metrics for portfolio strategies with Black-Litterman and MVO approaches.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
bl_lns_2	-0.025	-0.025	-21.868	0.517
bl_lo_2	0.768	1.557	-8.693	0.578
bl_lns_3	-0.038	-0.047	-30.982	0.512
bl_lo_3	0.911	3.082	-8.209	0.578
bl_lns_4	0.415	1.216	-10.510	0.540
bl_lo_4	1.210	3.492	-6.286	0.607
mvo_lns_2	0.231	0.304	-19.428	0.536
mvo_lo_2	1.042	2.132	-7.844	0.616
mvo_lns_3	0.118	0.199	-17.689	0.521
mvo_lo_3	1.091	3.558	-7.721	0.607
mvo_lns_4	0.418	0.444	-17.344	0.555
mvo_lo_4	1.092	3.383	-7.633	0.611
ew	0.406	0.587	-34.236	0.582

Note: Bold values indicate the best performance per column.

and estimation risk, may cause the two optimization methods to closely follow each other in performance.

The most robust and consistent performance is achieved by regime-aware ridge models. Table 13 and Figure 24 highlight the strong results of these models in all metrics evaluated, except for the Sortino ratio. The linear-ridge_lo_4 configuration attains the highest Sharpe ratio (1.341), while linear-ridge_lns_4 achieves the highest percentage of positive return days (0.654), underscoring the effectiveness of combining regularization with informative regime features. In particular, these regime-augmented ridge models outperform all MVO and EW benchmarks, not only in terms of average performance but also in terms of drawdown control and return stability. The cumulative return paths in figure 24 further emphasize the advantages of ridge models when enhanced with regime information. The best-performing ridge portfolio

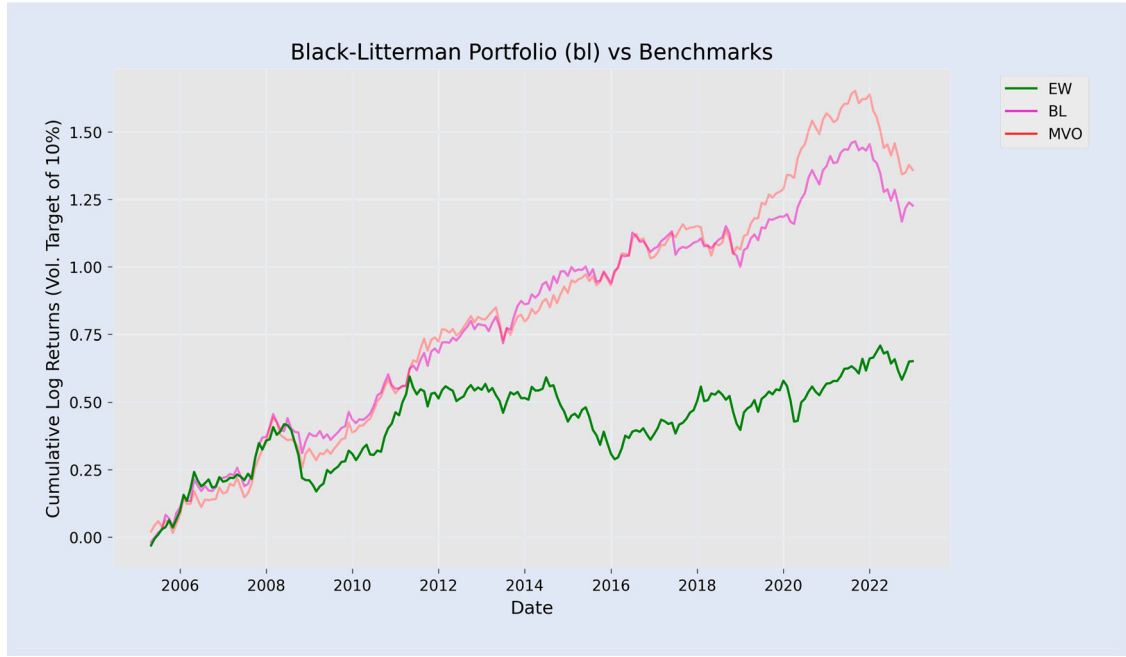


Figure 23. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Black-Litterman strategies are in pink, and mean-variance optimization strategies are in red.

Table 13. Performance metrics for portfolio strategies with ridge regression and MVO approaches. Bold values indicate the best performance per column.

Model	Sharpe	Sortino	MaxDD	% Positive Ret.
linear-ridge_lns_2	1.022	2.705	− 8.656	0.635
linear-ridge_mx_2	1.052	1.507	− 11.528	0.626
linear-ridge_los_2	0.900	1.471	− 9.568	0.621
linear-ridge_lo_2	0.762	1.100	− 11.528	0.583
linear-ridge_lns_3	1.287	2.571	− 6.520	0.645
linear-ridge_mx_3	1.133	1.876	− 5.385	0.597
linear-ridge_los_3	1.268	2.216	− 6.792	0.640
linear-ridge_lo_3	0.815	1.321	− 6.448	0.559
linear-ridge_lns_4	0.986	0.945	− 13.748	0.654
linear-ridge_mx_4	0.834	2.535	− 4.078	0.597
linear-ridge_los_4	1.341	2.058	− 6.379	0.640
linear-ridge_lo_4	0.637	1.790	− 6.645	0.564
mvo_lns_2	0.231	0.304	− 19.428	0.536
mvo_lo_2	1.042	2.132	− 7.844	0.616
mvo_lns_3	0.118	0.199	− 17.689	0.521
mvo_lo_3	1.091	3.558	− 7.721	0.607
mvo_lns_4	0.418	0.444	− 17.344	0.555
mvo_lo_4	1.092	3.383	− 7.633	0.611
ew	0.406	0.587	− 34.236	0.582

consistently outperforms both benchmarks across nearly the entire sample, exhibiting particularly strong resilience during MVO and EW drawdowns in 2008 and 2018.

Overall, these results highlight the broad applicability of regime-based enhancements across modeling approaches. While the magnitude of improvement varies, the consistent direction of gains underscores the value of macro-informed signals in guiding tactical allocation. Notably, the most substantial benefits emerge from simple, long-only strategies with low dimensionality, reinforcing the earlier insight that parsimony often outperforms complexity in regime-aware portfolio design.

7. Conclusion and future work

Using machine learning clustering techniques, we have developed a novel data-driven approach to regime detection using macroeconomic data. We demonstrate how it uncovers insights related to market dynamics and utilize our regime detection mechanism to derive a regime transition graph that aligns well with existing economic intuitions. Lastly, we demonstrate how these regimes can be utilized to enhance forecasting models and decision criteria by incorporating a probabilistic condition into historical data.

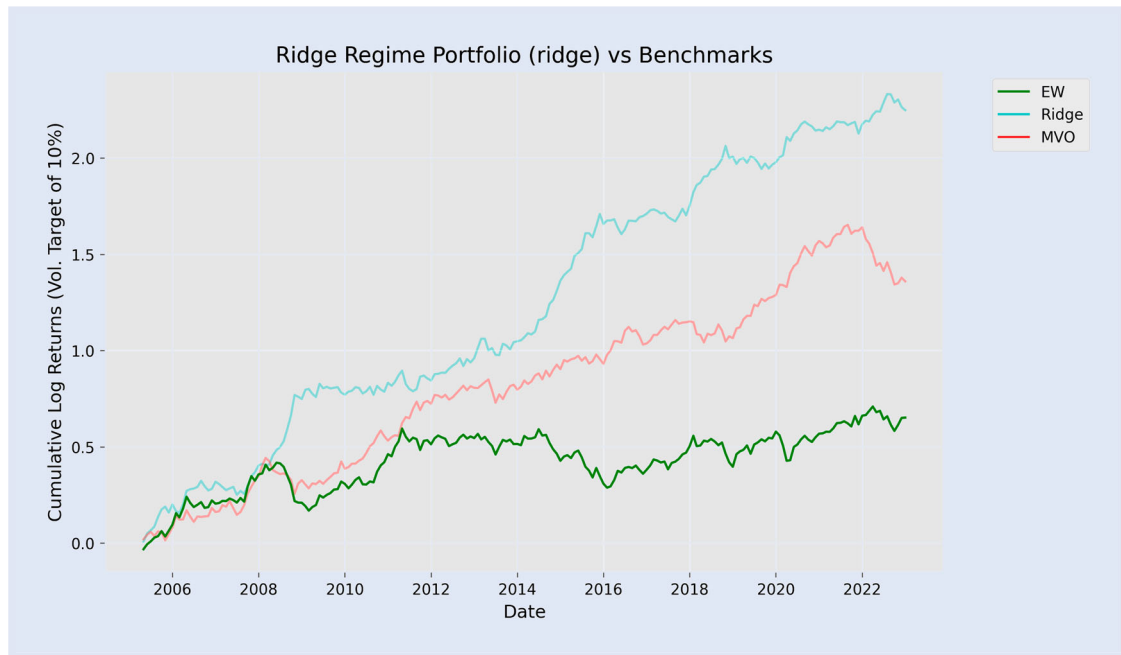


Figure 24. Cumulative log returns with 10% volatility scaling of various tactical allocation methods over time. Linear ridge strategies are in blue, and mean-variance optimization strategies are in red.

Further research on this topic could explore how more sophisticated clustering techniques can be applied to financial regime detection. The question of how the choice of macroeconomic characteristics affects the quality of the resulting cluster classifications remains. Finally, further work in this direction could also examine how these regimes could be used to improve other financial analytic processes.

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Data availability statement

The data supporting the findings of this study are openly available on the Federal Reserve Bank of St. Louis website following their work at <https://doi.org/10.20955/wp.2015.012> introduced by McCracken and Ng (2016).

ETF returns used for forecasting were calculated (or derived) based on data from Wharton Research Data Services (WRDS).

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

D. C. Oliveira <https://orcid.org/0009-0002-8845-8376>

D. Sandfelder <https://orcid.org/0000-0003-2164-4086>

A. Fujita <https://orcid.org/0000-0002-7756-7051>

X. Dong <https://orcid.org/0000-0002-1143-9786>

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